

3. Basic measurement in Epidemiology

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Content

- ▣ Measures of occurrence of events frequency (counts, ratios, proportions, rates)
- ▣ Measures of morbidity (prevalence and incidence rates)
- ▣ Measures of disease burden (DaLY, QUALY)
- ▣ Measures of mortality
- ▣ Standardization of rates
- ▣ Measures of disease occurrence

Why we measure disease frequency ?

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It is fundamental to epidemiology

- To study distribution of disease
- To make comparisons.
- For monitoring the health status of the population
- For planning health services

How do we measure diseases?

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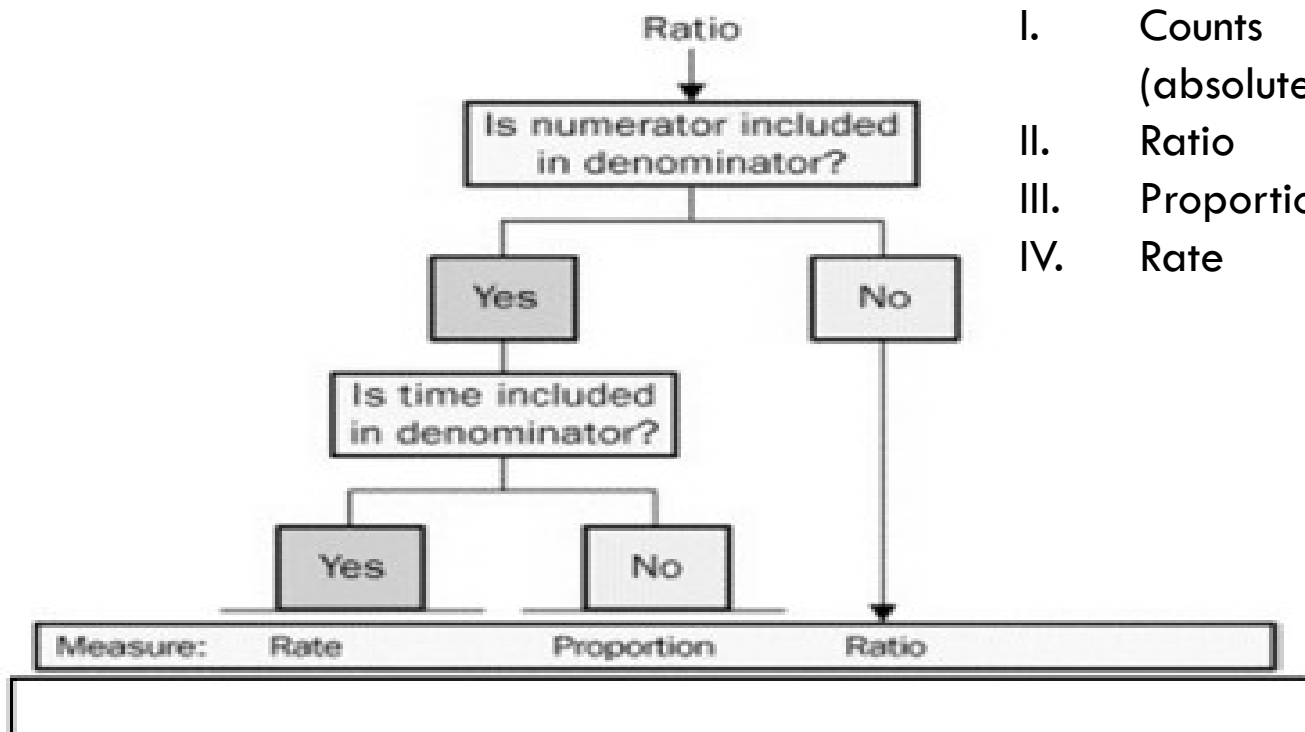
- Four quantitative descriptors
 - ▣ Numbers , Ratios , Proportions , Rates
- Counting case, events, phenomena (*numbers*)
 - ▣ *Numbers*: Use of actual number of events
- E.g. 100 cases of TB in community A

Algorithm for distinguishing rates, proportions, and ratios

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Four quantitative Tools in Measures of frequency

- I. Counts
(absolute_numbers)
- II. Ratio
- III. Proportion
- IV. Rate



Simple Count(absolute numbers)

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- Number of cases in a defined population
- May be sufficient by itself for some purposes e.g. to determine the quantity of health care resources required to meet needs for people who currently have the disease.
 - Can be used to compare disease frequency between populations if and only if the population are known to be of similar size.
 - Often hard to interpret if populations being compared *differ in size*
 - hence counts are sometimes termed “numerator data”.

Measurement

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Ratio

- The value obtained by dividing one quantity by another.
- Used to compare two quantities both descriptive measures and as analytic tools
- The entities represented by the two numbers are not required to be related to one another.
- In other words, the individuals in the numerator can be different from those in the denominator.
- For example, the “sex ratio” is a ratio of two unrelated numbers
- Usually expressed as the number of males per 100 females.
- Child-women ratio
- e.g. Ratio of TB cases in community A to B is 1:10
- Sex ratio
- **Fetal death *ratio*: Fetal deaths/live births.**
- # PG students per lect.

Ratio= **N/D**

Measurement...

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Proportion

- The ratio of a part to the whole
 - Is also one number divided by another, but the entities represented by
 - ▣ These numbers are related to one another.
 - In fact, the numerator of a proportion is always a subset of the denominator.
 - Proportions, also known as decimal /common fractions, are often expressed as percentages and range from 0 to 1 or 0% to 100%.
 - e.g. proportion of diabetics in a population
 - Fetal death *rate*: Fetal deaths/all births
- All births includes both live births and fetal deaths.
- =N/D (prevalence, case fatality of an event)

Measurement...

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□ Rate

- Is also one number divided by another, but **time** is an integral part of the denominator.
- We are familiar with rates in our daily travels since a rate is a measure of how fast we travel.
- For example, the maximum speed or rate at which cars are permitted to travel may be 65 miles per hour.
- This rate can also be written as 65miles/one hour. The measure of time in the denominator is what makes this number a rate.
- It has five components(numerator, denominator, time, population, and multiplier)
- Measles cases in under five children in 2000 x k
Under five children in 2000

$$\text{Rate} = \frac{\text{Number of events in a specfied population during a specfied period}}{\text{Average popluation during the period}} * 10^n$$

Characteristics of simple frequency measures

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- ◇ All rates are proportions.
- ◇ All rates are ratios too.
- ◇ All proportions are ratios.
- ◇ Not all proportions are rates.
- ◇ But not all ratios are proportions.

	Incidence	Prevalence
Numerator	Number of new cases of disease during a specified period of time	Number of existing cases of disease at a given point of time
Denominator	Population at risk	Population at risk
Focus	Whether the event is a new case Time of onset of the disease	Presence or absence of a disease Time period is arbitrary; rather a "snapshot" in time
Uses	Expresses the risk of becoming ill The main measure of acute diseases or conditions, but also used for chronic diseases More useful for studies of causation	Estimates the probability of the population being ill at the period of time being studied. Useful in the study of the burden of chronic diseases and implication for health services

Note: If incident cases are not resolved, but continue over time, then they become existing (prevalent) cases. In this sense, prevalence = incidence × duration.

Measurement...

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Types of rates

- A. Crude rates
- B. Specific rates
- C. Adjusted (standardized) rates: It is a...
 - ◇ **Classic Epidemiological Method for the Comparison of Rates.**

Crude rate

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- Accounts the total burden of health outcomes in all ages and cause.
- Are overall measure that does not take explicit account of the composition of the population.
- Its value will be an average of the values for the individual subgroups, weighted by their relative sizes [Democracy].
- Is the simplest and most straightforward summary.
- $CR = \frac{E}{P} * K$ (E= Number of events, p=mid-year total population, k=constant, may be 100, 1000, etc).
- Are summary rates based on the actual number of events (births, deaths, diseases) in the total population over a given time period
- Crude birth rate (CBR) and the crude death rate (CDR) are widely used
- CDR depend on two factors
 - The probability of dying for individuals
 - The age distribution of the population

Crude...

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➤ **Advantages**

- Actual summary rates
- Calculable from minimum information
- Widely used despite limitations

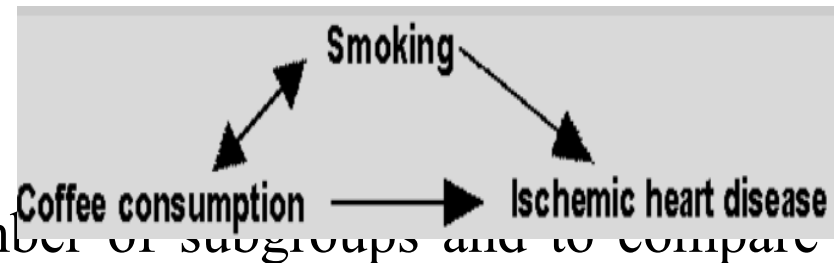
➤ **Disadvantages**

- Difficult to interpret due to variation in composition (e.g. age)
- Obscure significant differences in risk between subgroups

Specific rates

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- Apply to specific subgroups in the population.
- **SR** are calculated based on the specific characteristic(s).
- Denominator should be the population in that specific group except for **cause-specific rates**
- **Advantages**
 - The rates apply to homogenous subgroups
 - The rates are detailed and useful for epidemiological and public health purposes.
 - Important to control third variables
- **Disadvantages**
 - Impractical to work with a large number of subgroups and to compare many of two or more populations
 - Furthermore, if the subgroups consist of small populations, the specific rates can be very imprecise.



Category of Specific Rates

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❑ Calculation of factor specific disease frequency measures such as:

- ✓ Cause specific rates
- ✓ Age specific incidence rates
- ✓ Age specific death rate
- ✓ Sex-specific death rates
- ✓ Race-specific rates
- ✓ Place-specific rates

Adjusted/standardization rates

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- Epidemiologists are always mindful of population diversity.
- Is a way to make fairer comparison between groups with different age distribution.
- Different FD in different populations, a comparison between CR would be misleading since CRs are not very informative about the health status of a population.
- Standardization for the characteristic(s) responsible for the differences in comparison is necessary.
- Age and sex are two of the most common variables used for standardization and they are called standardized rates.
- The difference between CR and SR is that the former are calculated based on the population under study as a whole whereas the latter are based on particular characteristic(s) as standard.

Adjusted/standardization rates...

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- A principal role in epidemiology is to compare the incidence of disease or mortality between two or more populations.
- However, the comparison of crude mortality or morbidity rates is often misleading because the populations being compared may differ significantly with respect to certain underlying characteristics, such as age or sex, that will affect the overall rate of morbidity or mortality
- For example, age is an important determinant of mortality.
- An older population will have a higher overall mortality rate than a younger population.
- As a result, variations in age will complicate any comparison between two or more populations that have different age structures.

Adjusted/standardization rates...

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Category specific rates

- One method of overcoming the effects of confounding variables such as age is to simply present and compare the age specific rates.
- While this allows for a more comprehensive comparison of mortality or morbidity rates between two or more populations, as the number of stratum specific rates being compared increases, the volume of data being examined may become unmanageable.
- It is, therefore, more useful to combine category specific rates into a single summary rate that has been adjusted to take into account its age structure or other confounding factor.
- **This is achieved by using the methods of standardisation**

Adjusted/standardization rates...

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- Virtually every large population is heterogeneous in regard to
 - ▣ Socio-demographic (e.g., age, gender, education, religion), geographic, genetic, occupational, dietary, medical history, and
 - ▣ Numerous other personal attributes and
 - ▣ Environmental factors related to health.
- The adjusted or “standardized” rate is obtained by dividing the total of expected cases by the standard population

Adjustment...

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- In many circumstances, it is useful to have single summary rate for each population that takes into account any differences in the structure of the population
- This is done through **adjustment** or **standardization**
- Adjusted rates are summary rates that have undergone statistical transformation, to permit fair comparison between groups differing in some characteristics that may affect risk of disease.

Measurement...

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- When comparing the crude death rates of two or more places, it is impossible to know whether the difference is due to age composition, age specific death rate or both.
- In age adjusted rates the difference is exclusively attributed to differences in age specific mortality rates, since the effect of age composition is artificially removed.

Goals of Adjustment Procedures

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- ❖ Appropriately **combine** data
- ❖ Make appropriate **comparisons** among groups
- ❖ Reach appropriate **conclusions** and inferences
- ❖ There are two methods for calculating standardized rates, namely
- ❖ **Direct and indirect standardization**
- ❖ The ratio of the observed number of deaths to the expected number of deaths is called: “Standardized mortality ratio” or SMR.

$$\text{Directly standardized rate} = \frac{\sum (\text{stratum-specific rates} \times \text{standard weights})}{\sum (\text{standard weights})}$$

$$\text{Directly standardized rate} = \frac{(r_1 N_1 + r_2 N_2 + r_3 N_3 + \dots + r_n N_n)}{(N_1 + N_2 + N_3 + \dots + N_n)}$$

$$R_s = \frac{\sum (r_k \times N_k)}{\sum (N_k)} = \sum \left(r_k \times \frac{N_k}{\sum (N_k)} \right) = \sum \left(r_k \times \frac{N_k}{N} \right)$$

$$R_s = \sum (r_k W_k)$$

where:

r_k = rate in k-th stratum of the *study* population

N_k = number of persons in k-th stratum of the *standard* population

N = total number of persons in the *standard* population ($\sum N_k$)

W_k = weight for each stratum (equal to N_k/N)

$\sum$ means summation over the k strata.

Standardization

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- The 3's (Standardization, simplification, specifications.
- Applicable for factors other than age(socioeconomic status, income, race, are of residence, etc.)
- The adjusted or “standardized” rate is **obtained by dividing the total of expected cases by the standard population.**

Direct standardization

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- Used when specific mortality rates for two or more populations are known such as age(ASMR).
- Steps
 - A. Calculate the age-specific mortality rates for each age group in each population
 - B. Then choose the standard (reference) population from one of the populations like national mortality
 - C. Multiply the age-specific mortality rates of the other population under study to the number of persons in each age group of the standard population

Direct standardization...

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- D. By this way, you will get the expected deaths for each age group of each population(c)
- D. Add the number of expected deaths from all age groups.
- D. Finally to get the age-adjusted mortality rates, divide the total number of expected deaths by the standard population
- D. Now you can conclude by comparing the age-standardized mortality rates of two populations

Adjusted rate...

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➤ Advantages

- Are summary rates
- Permit unbiased comparison
- Easy to interpret

➤ Disadvantages

- Ineffectual rates
- Absolute magnitude depends on standard population
- Opposing trends in subgroups masked

Direct standardization: steps WITH example

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- Standardized rates are **a statistical measure of any rates in a population.**
- These are adjusted rates that take into account the vital differences between populations that may affect their birthrates or death rates.
- Traditionally, there are two methods of standardization: **direct and indirect standardization.**
- Population of interest ASMR^k Standard population
→ compare Age Adjusted MR

Methods of Standardisation

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- There are two methods of standardisation commonly used in epidemiological studies, and these are characterized by whether the standard used is a population distribution (direct method) or a set of specific rates (indirect method).
- Both direct and indirect standardisation involves the calculation of numbers of expected events (e.g. deaths), which are compared to the number of observed events.
- Age is a factor that is frequently adjusted for in epidemiological investigations, particularly in comparative mortality studies, since the age structure of a population will greatly affect the population's overall mortality.
- To illustrate the methods of both direct and indirect standardisation, the age specific mortality rates for two hypothetical populations are compared below in Ethiopia.

Direct method of standardisation

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	Country A			Country B		
Age - group	No. of deaths	Population	Rate per 1,000 pyrs	No. of deaths	Population	Rate per 1,000 pyrs
0-29	7,000	6,000,000	1.2	6,300	1,500,000	4.2
30-59	20,000	5,500,000	3.6	3,000	550,000	5.5
60+	120,000	2,500,000	48	6,000	120,000	50
Total	147,000	14,000,000	10.5	15,300	2,170,000	7

presents CMR
The overall CMR is higher for country A (10.5/1,000 person years) but the ASMR higher among all age-groups in country B.

For Standard population of the above three age groups 100 thu; 65 thu; and 20 thu, the Expected Deaths as

	Country A	Country B
	Expected deaths	Expected deaths
0-29	$0.0012 \times 100,000 = 120$	$0.0042 \times 100,000 = 420$
30-59	$0.0036 \times 65,000 = 234$	$0.0055 \times 65,000 = 357.5$
60+	$0.048 \times 20,000 = 960$	$0.05 \times 20,000 = 1,000$
Total expected deaths	1,314	1,777.5
Age adjusted rate	$1,314/185,000 = 7.1$ per 1,000 pyrs	$1,777.5/185,000 = 9.6$ per 1,000 pyrs
Age standard rate ratio (B:A) = $9.6/7.1 = 1.35$		

Indirect method of standardisation

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- The indirect method of standardisation is commonly used when age-specific rates are unavailable.
- For example if we did not know the age specific mortality rates for country B
- In this method, instead of taking one population structure as standard and applying sets of rates to it to estimate expected events, a set of rates from a standard population (country A) is applied to each of the populations being compared to calculate standardized morbidity/mortality ratios.
- Indirect method of standardisation: Number of expected deaths if the population had the same age-specific mortality rates as Country A.

Indirect method of standardization...

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	Country A	Country B
	Expected deaths	Expected deaths
0-29	$0.0012 \times 6,000,000 = 7,200$	$0.0012 \times 1,500,000 = 1,800$
30-59	$0.0036 \times 5,500,000 = 19,800$	$0.0036 \times 550,000 = 1,980$
60+	$0.048 \times 2,500,000 = 120,000$	$0.048 \times 120,000 = 5,760$
Total expected deaths (E)	147,000	9,540
Total observed deaths (O)	147,000	15,300
Standardised Mortality Ratio O/E x 100	100	160

- The IMS is used to calculate how many deaths would be expected in Country B if it had the same age-specific mortality rates as Country A.
- The expected deaths in Country B are calculated by multiplying the age specific rate for Country A by the population of Country B in the corresponding age group.
- The sum of the age categories gives the total number of deaths that would be experienced in country B if it had the same mortality experience as country A.
- An overall summary measure can then be calculated, that is, the standardized mortality ratio (SMR), which is the ratio of the observed number of deaths to the expected number of deaths.
- $SMR = \frac{\text{Observed number of deaths (O)} \times 100\%}{\text{Expected number of deaths (E)}}$
- $SMR = \frac{160}{100} = 1.6 \times 100 = 160$, which means that the number of observed deaths in Country B is 60% higher than Expected

The crude mortality rate for all ages in the United States for 1995-1997 is 8.7 per 1,000 population. In Mexico it is much lower: 4.7 per 1,000 population.

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Calculate

- The adjusted rate of the two countries(DSR)

	Standard population (1)	Age-specific mortality rate per 100,000 population, 1995-1997		Expected number of deaths	
		Mexico (2)	United States (3)	Mexico (4)=(1)x(2)/100,000	United States (5)=(1)x(3)/100,000
<1	2,400	1693.2	737.8	41	18
1-4	9,600	112.5	38.5	11	4
5-14	19,000	36.2	21.7	7	4
15-24	17,000	102.9	90.3	17	15
25-44	26,000	209.6	176.4	55	46
45-64	19,000	841.1	702.3	160	133
65+	7,000	4,967.4	5,062.6	348	354
	100,000			639	574

- When eliminating the effect of the difference in the age structure in both countries, we obtain a rate that is higher in ---than in the ---.
- Is the conclusion of the comparison of the rates is inversed or not when using adjusted rates rather than crude rates.

Table 1
Population and Deaths by Age in 1970 for White Females in
Miami, Alaska, and the U.S.

Age	Miami			Alaska			U.S.		
	Pop.	Deaths		Pop.	Deaths		Pop. ⁺	Deaths ⁺	
< 15	114,350	136		37,164	59		23,961	32	
15-24	80,259	57		20,036	18		15,420	9	
25-44	133,440	208		32,693	37		21,353	30	
45-64	142,670	1,016		14,947	90		19,609	140	
65+	92,168	3,605		2,077	81		10,685	529	

EX1. Home take exercise

Calculate the CDR and SAR Of Each State?

Indirect standardization (IS)

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- Common practice is to compare specific mortality rate indirectly
 - A. Choose a reference or standard population.
 - B. Calculate the observed number of deaths in the population (s) of interest
 - C. Apply the age-specific mortality rates from the chosen reference population to the population(s) of interest.
 - D. Multiply the number of people in each age group of the population(s) of interest by the age-specific mortality rate in the comparable age group of the reference population.
 - E. Sum the total number of expected deaths for each population of interest.
 - F. Divide the total number of observed deaths of the population(s) of interest by the expected deaths

Measures of morbidity rate

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- Morbidity rates are rates that are used to quantify the magnitude/frequency of **diseases/illness**.
- The frequency of health related events are measured by risk, prevalence and incidence rate.
- Two common morbidity rates
 - ❖ **Incidence rates (Cumulative incidence, incidence density)**
 - ❖ **Prevalence (Period prevalence, point prevalence)**

Incidence {Risk}

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- The occurrence of new cases of disease that develop in a candidate population **over a specified time** period (measure of absolute risk).
- Measures the newly diagnosed patients develop over time in [Week, month, years so on]
- If the IR of a certain disease is high in one area, then the risk of acquiring that disease by other healthy individuals will be high.

THREE KEY IDEAS

- Incidence measures new disease events.
- New cases of disease are measured in a candidate/risk/ population
- Account the specific amount of time (transition from health to disease).
- Example. In Sept 2019 there were 50 new cases of relapsing fever in “Kebele X”. The average total population of “Kebele X” was 5000. Calculate the incidence rate of relapsing fever in “Kebele X” in Ginbot 2019.
- The types of incidence rate is attack rate (used during outbreak)
- Used for etiological study

Risk(cumulative incidence-CI)

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- Likelihood that an individual will contract a disease.
- The proportion of a candidate **population{closed}** that becomes diseased over a specified period of time.

Mathematically

$$\text{Risk} = \frac{\text{New cases occurring during a given time period} * k}{\text{Population at risk during the same period}}$$

- Numerator (new cases of disease) is subset of denominator (candidate population).

CI...

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- The average risk of getting a disease over a certain period of time.
- A risk is the probability of getting a disease.
- CI is mainly used in fixed populations when there are no or small losses to follow-up.
- Possible value ranges from 0 to 1 or 0% to 100%
- Time is not integral part but expressed by words that accompany by the CI number.
- Thus CI is dimensionless
- **For example: -**
- If we count all new cases of influenza occurring in X undergraduates from September 1, 2018 – October 28, 2019, and we take as the denominator all undergraduates enrolled in September 1, 2018, we would be describing the **cumulative incidence rate** of influenza.

Incidence density/person-time/PTY

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- The risk/probability of an individual developing a disease overtime {closed or open Population}
 - The rapidity with which new cases of a disease develop overtime
 - Occurrence of new cases of disease that arise during *person time* of observation.
- Incidence Rate =
$$\frac{\text{Number of new cases of disease}}{\text{Person-time of observation in candidate population}}$$
- The difference between
 - The incidence rate's denominator integrates time (t), and so it is a true rate.
 - Incidence rate only up until he or she is diagnosed with the disease of interest.
 - Unlike for cumulative incidence, Person-time is accrued only while the candidate is being followed.

IR...

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- Other things being equal, the number of disease events observed in a population over times varies directly with the **number of people at risk and the amount of time** over which they are observed.
- Is a fundamental tool for etiologic studies of diseases since it is a direct measure of risk.
- If the IR is high in a specific area , the event must be systematically searched.
- Person-time rates are often used in cohort (follow-up) studies of diseases with long incubation or latency periods, such as occupationally related diseases, AIDS, and chronic diseases.
- **ATTACK RATE:** The proportion of a group that experiences the outcome under study over a given period (e.g., the period of an epidemic)
- **SAR:** The number of cases of an infection that occur among contacts within the incubation period following exposure to a primary case in relation to the total number of exposed contacts
- The SAR is a measure of contagiousness and is useful in evaluating control measures.

Why PTY/ID?

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- ❖ Impossible to follow everyone for entire time period.
- ❖ In a dynamic population, individuals enter population **over different times**, become lost, etc.
- ❖ Length of follow-up is not uniform for all .
- ❖ Incidence rates do not make assumption of complete follow-up.
- ❖ **Since the number of cases is divided by a measure of time of observation, rather than people, this helps address the problem of competing risks.**
- ❖ **CR:** An event that removes a subject from being at risk for the outcome under investigation.
- ❖ For example, in a study of smoking and cancer of the lung, a subject who dies of CHD is no longer at risk of lung cancer; in this situation, CHD is a competing risk.

Example

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- Investigators enrolled 2,100 men in a study and followed them over 4 years to determine the rate of CHD.
 - We assume that persons diagnosed with disease and those lost to follow-up were disease-free for half of the year, and thus contribute $\frac{1}{2}$ year to the denominator.
 - Initial enrolment: 2,100 men free of disease
 - ▣ After 1 year: 2,000 disease-free, 0 with disease, 100 lost to follow-up
 - ▣ After 2 years: 1,900 disease-free, 1 with disease, 99 lost to follow-up
 - ▣ After 3 years: 1,100 disease-free, 7 with disease, 793 lost to follow-up
 - ▣ After 4 years: 700 disease-free, 8 with disease, 392 lost to follow-up
1. Identify x : $x = \text{cases diagnosed} = 1 + 7 + 8 = 16$
 2. Calculate y , the person-years of observation:

Results

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$$(2,000 + \frac{1}{2} \times 100) + (1,900 + \frac{1}{2} \times 1 + \frac{1}{2} \times 99) + (1,100 + \frac{1}{2} \times 7 + \frac{1}{2} \times 793) + (700 + \frac{1}{2} \times 8 + \frac{1}{2} \times 392) = 6,400 \text{ person-years of observation.}$$

A second way to calculate the person-years of observation is to turn the data around to reflect how many people were followed for how many years, as follows:

700 men x 4.0 years = 2,800 person-years

8 + 392 = 400 men x 3.5 years = 1,400 person-years

7 + 793 = 800 men x 2.5 years = 2,000 person-years

1 + 99 = 100 men x 1.5 years = 150 person-years

$$\begin{aligned} 3. \text{ Person-time rate} &= \frac{\text{number of cases during 4-year study}}{\text{time each person was observed, totaled for all persons}} \times 10^n \\ &= \frac{16}{6,400} \times 10^n = .0025 \times 10^n \end{aligned}$$

or, if 10^n is set at 1,000, there were 2.5 cases per 1,000 person-years of observation. This quantity is also commonly expressed as 2.5 cases per 1,000 persons per year.

Practical challenges in measuring incidence

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- I. Identification of population at risk
- II. Population is not static/it fluctuates/as a result of births, deaths and migration
- III. People are at risk only until they get the disease and then no more at risk

I. Use the total population as a denominator

- This gives an estimate of the incidence rate and not the actual incidence rate

II. Use person-time at risk

- Incidence density = number of new cases of a disease over a specified period / person-time at risk

Prevalence {Burden}

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- ▣ Measures the existence of current disease.
 - Didn't provide strong evidence of causality.
 - Has three components numerator, denominator, and multiplier.
 - Proportion of the total population that is diseased during a specified period or at a point in time

$$\text{Prevalence} = \frac{\text{Number of people with the disease/conditions at specified time}}{\text{Number of people in the population at risk at the specified period}} * 100$$

- ▣ **Types**
 - ▣ **Point prevalence rate**
 - ▣ **Period prevalence rate**
 - ▣ **Lifetime prevalence**

Point PR

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- Measures the **proportion** of a population with a disease at a point in time
- Determined by conducting cross-sectional/survey study.
- Point prevalence rate=All persons with a disease at a point in time/Total population
- proportion of the population that is diseased at a single point in time and can be thought of as a single snapshot of the population.
- It is not a rate, but a true proportion
- e.g. December 1, 2010 or a point in someone's life such as college graduation.
- **Point prevalence** =
$$\frac{\text{Number of existing cases of disease}}{\text{Number in total population}}$$
 At a point in time

Period PR

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- Measures the proportion of a population with a disease in a specified time period
 - Period prevalence rate = $\frac{\text{All persons with a disease overtime period}}{\text{Average(mid-year)population in the same period.}}$
 - e.g. during the year 2010.
 - The period prevalence includes the number of cases that were present at the start of the period.
 - e.g. January 1, 2010 as well as the number that developed during the period (for example, January 2 through December 31, 2011).
- E.g. annual prevalence rate/ lifetime prevalence rate.
- A series of snapshots $\frac{\text{Number of existing cases of disease}}{\text{Number in total population}}$ During a period of time
 - **Period prevalence** =

Example

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Two surveys were done in the same community 12 months apart. Of 5,000 people surveyed the first time, 25 had antibodies to histoplasmosis. Twelve months later, 35 had antibodies, including the original 25. Calculate the prevalence at the second survey, and compare the prevalence with the 1-year incidence.

1. Prevalence at the second survey:

x = antibody positive at second survey = 35

y = population = 5,000

$$x/y \times 10^n = 35/5,000 \times 1,000 = 7 \text{ per } 1,000$$

2. Incidence during the 12-month period:

x = number of new positives during the 12-month period = $35 - 25 = 10$

y = population at risk = $5,000 - 25 = 4,975$

$$x/y \times 10^n = 10/4,975 \times 1,000 = 2 \text{ per } 1,000$$

Characteristics of Prevalence

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- Cause and effect measured simultaneously
 - ▣ Impossible to infer causation
- Useful for planning (e.g. beds, clinics, workforce needs)
- High prevalence $\neq$ high risk
 - ▣ could reflect increased survival(improved care, behavior change - long duration)
- Low prevalence $\Rightarrow$ could reflect rapid fatal or cure process - short duration)
- Easy to obtain $\Rightarrow$ need only 1 measurement

Measure of mortality/death rates

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- These rates measures magnitude of **deaths** in a community
 - ▣ Some are crude like the crude death rate
 - ▣ Others are cause-specific mortality rate
 - ▣ Some others are adjusted like standardized mortality ration
- Rates are calculated using mid-interval/average population due to population fluctuation over time.
- **Below are given some formulas for the commonly used mortality rates and ratios.**

Common Mortality rates

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- Crude death rate
- Age-specific mortality rate
- Sex-specific mortality rate
- Cause-specific mortality rate
- Proportionate mortality ratio
- Case fatality rate
- Fetal death rate
- Perinatal mortality rate
- Neonatal mortality rate
- Infant mortality rate
- Child mortality rate
- Under-five mortality rate
- Maternal mortality ratio

5. Epidemiological Study designs

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- Descriptive study designs
 - ▣ Ecological/correlation
 - ▣ Case reports/case series
 - ▣ Cross-sectional
- Analytical study designs:
 - ▣ Case-control
 - ▣ Cohort
 - ▣ Experimental

PURPOSE

- To test hypothesis about causal relationship
- To search for cause & effect
- To compare treatment regimens / prevention programs
- To assess diagnostic tests
- To quantify the association b/n exposure & outcome (measure of association)
- Focuses on determinants of diagnosis by testing hypothesis

Basic Question in Analytic Epidemiology

55

- Are exposure and disease linked?

- Exposure  Disease

- Look to link exposure and disease

- What is the exposure?
 - Who are the exposed?
 - What are the potential health effects?
 - What approach will you take to study

- The relationship between exposure and effect?

Prospective Study

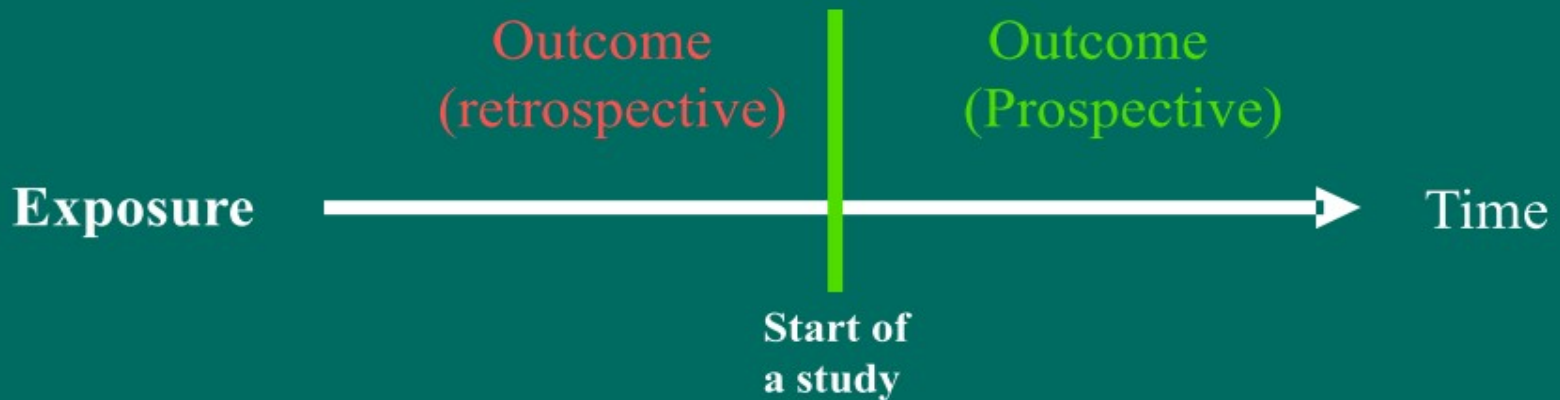
- ▣ looks forward,
- ▣ looks to the future, examines future events, follows a condition, concern or disease into the future

Retrospective Study

- ▣ to look back',
- ▣ looks back in time to study events that have already occurred

Prospective Vs Retrospective Study

- In prospective study outcome has not occurred at the beginning of the study.*



Types of analytic study design

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There are two types of analytic study designs

- ▣ Observational- to simply observe
- ▣ Interventional - manipulation or determination

OBSERVATIONAL STUDIES

- ▣ Information is obtained by simple observation of the event (two basic types): -
 - ▣ Case control study
 - ▣ Cohort study
 - ▣ Cross sectional (some times)

COHORT STUDIES- Definition

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- Cohort
 - ▣ A group of people who share a common experience or condition
 - ▣ E.g. Birth cohorts, cohort of smokers, occupational exposures
- Cohort studies
 - ▣ The observation of a cohort over time to measure outcome(s)
 - ▣ Because the data on **exposure and disease** refer to different points in time, cohort studies are longitudinal, like case-control studies
 - ▣ Longitudinal, follow-up or incidence studies , **prospective study??**

Definition ...

60

- They have 2 primary purposes:
 - ▣ **Descriptive:** to describe the incidence rates of an outcome
 - ▣ **Analytic:** to analyze associations between the outcomes and risk factors
 - Usual type
- Begin with a group of people free of disease
- Who are classified into subgroups according to exposure to a potential cause of outcome
- The whole cohort is followed up to see how the subsequent development of outcome differs between the groups with and without exposure (Figure 1)

Definition...

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BASIC ELEMENTS

- Disease free at entry
- Selected by exposure status rather than outcome
- Follow up is needed to determine the incidence of the outcome
- Compares incidence rates of exposed against unexposed group

a. EXPOSURE STATUS

- Study subjects should be disease free.
- Define study subjects using inclusion /exclusion criteria

b. POSSIBLE OUTCOME IN COHORT

1. No disease
2. Diseased
- 3 Lost to follow up

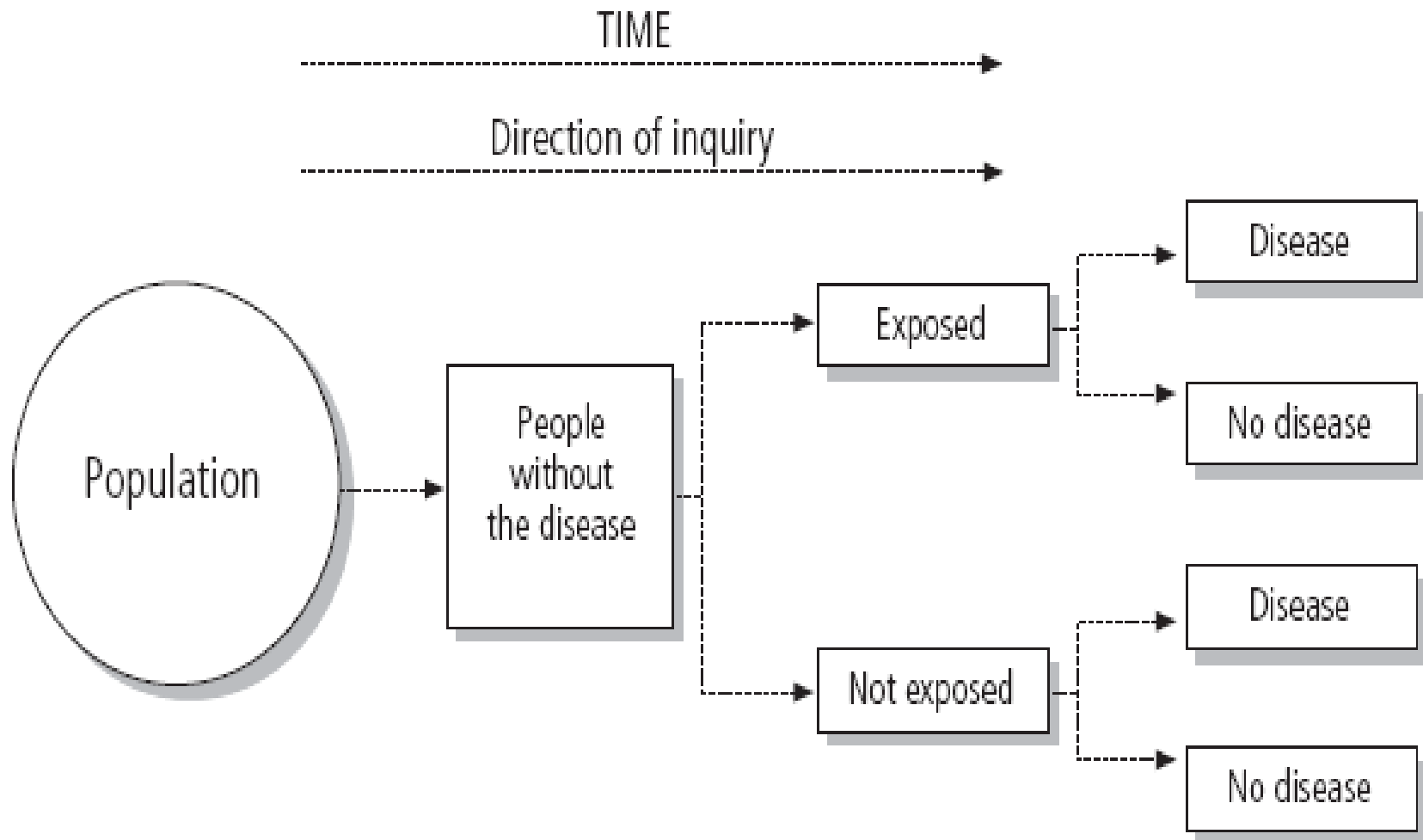


Figure 1- design of cohort study

Types

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- Prospective vs. Retrospective (Concurrent vs. Non-concurrent)
 - ▣ Depending on temporal relationship between **initiation of the study /timing of data collection/ and occurrence of the disease**
 - ▣ Cohort studies have been called **prospective studies**, but this terminology is confusing and should be avoided
 - the term “prospective” refers to the timing of data collection and not to the relationship between exposure and effect

Prospective cohort studies

64

- Exposure and outcome data is collected after start of the study
 - ▣ cohorts Identified in the present →
 - ▣ exposure status or possible explanatory/prognostic factors determined in the present →
 - ▣ Cohorts followed-up to identify outcome →
 - ▣ Ascertainment of outcome done in future →

Retrospective cohort

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- All the exposure and effect data have been collected before the actual study begins
- This type of investigation is called **a historical cohort study**
- Conduct
 - ▣ Identify cohort in the past using records/databases
 - ▣ Determine exposure or prognostic factors in the past using again records or databases then →
 - ▣ Identify outcome in past or present or future (in case of mixed cohort) →

Retrospective cohort...

66

- Costs can occasionally be reduced by using a historical cohort (identified on the basis of records of previous exposure)
- This sort of design is relatively common for studies of cancer related to occupational exposures
- For example, records of military personnel exposure to radioactive fall-out at nuclear bomb testing sites have been used to examine the possible causal role of fall-out in the development of cancer over the past 30 years

Retrospective vs. Prospective

67

- Can be conducted more quickly and cheaply
 - ▣ All relevant events have occurred
 - ▣ Efficient for disease with long latency periods
- Depend on availability of routine data
 - ▣ Incompleteness
 - ▣ Lack of data on confounding variables

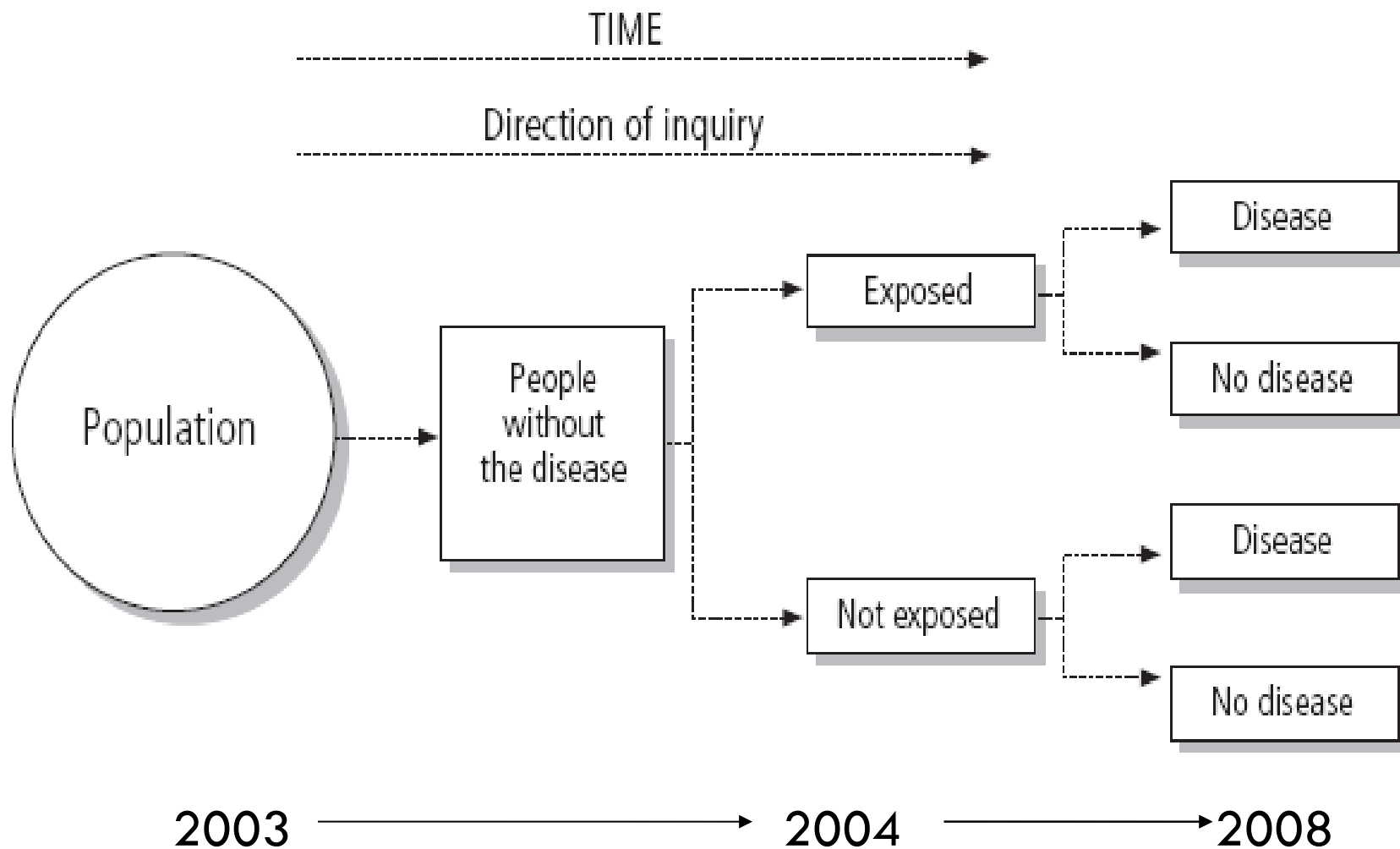


Fig. 3. Design of retrospective cohort studies

Design and data collection

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1. Define and identify cohorts

1.1. Identify population at risk

□ Selection of Exposed Population

- Depends on research question
- Outcome must not be rare in exposed
- Accessible and compliant subjects
 - E.g., Nurse's Health Study, Physicians

□ Selection of Non-exposed Group

- Similar to exposed
- Control for confounding factors

Design and data collection...

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- 1.2. Screen identified subjects for the disease and omit the prevalent cases
2. Define, assess, identify and classify exposure
3. Follow-up and ascertain outcome
 - ▣ Timing of outcome events-case definition

□ Self Report (interviews, surveys, etc.)

□ Advantages:

- Opportunity to question subjects on as many factors as necessary.
- Good for collecting information on exposures not routinely recorded.

□ Disadvantages:

- Subject to response bias (e.g. due to stigma, response expectations, etc.).
- Subject to interviewer bias

□ Pre-existing Records:

□ Advantages:

- **Inexpensive** and Relatively **easy** to work with
- Usually unbiased -data were collected for other purposes

Sources of Exposure Information...

72

□ **Disadvantages:**

- Exposure information may not be precise enough to address the research question.
- Records frequently do not contain data on potential confounding factors.

□ **Direct Measurement**

- Can provide objective and unbiased exposure ascertainment (e.g. blood pressure, serum samples, environmental measurements, etc.).

□ **Death certificates** (National Death Index) –for some causes

□ **Clinical history**

□ **Medical examination** (periodic re- examination of the cohort)

□ **Hospital discharge logs**

Sampling

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Sample size - for test of significant difference between two proportions, the following formula can be used:

$$n = \frac{(Z_{\alpha} + Z_{2\beta})^2 \{p_1(1 - p_1) + p_2(1 - p_2)\}}{(p_1 - p_2)^2}$$

Parameters:

n - size of sample in each group

P_1, P_2 —estimated population prevalence in the comparison groups

$\beta = 1 - \text{Power}$ (the probability that if the two proportions differ the test will produce a significant difference)

- ▣ Usually a power of 80% is used

- $RR = \text{INCIDENCE EXP} / \text{INC NONEXP}$
- Presence of association
 - $RR=1$ – no association; $RR<1$ – negative association; $RR>1$ – positive association
 - $AR \& AR\%$, $PAR \& \%$,
 - $P\text{-value} < 0.05$ – statistically significant association
- Strength of association
 - ▣ Weak – RR close to zero; Strong – RR far from zero

Advantages

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- Elucidates temporal relationship between exposure and outcome (It shows temporal sequence)
- it is possible to examine a range of outcomes of particular value when exposure is rare
- Minimize selection bias – **prospective**
- Can examine multiple effect of a single exposure
- Allow direct measure of incidence of disease in the exposed & none exposed.

Disadvantages

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- **Time-consuming - prospective**
 - ▣ Require long periods of follow-up since disease may occur a long time after exposure
 - ▣ Many exposures investigated are long-term in nature and accurate information about them requires data collection over long periods
- **Expensive - prospective**
 - ▣ Often involve following large number of individuals for many years
- **Incompleteness of records – retrospective**
 - ▣ Require the availability of adequate records
- **Loss-to-follow-up – bias**
 - ▣ Validity of results - seriously affected by losses to follow up
- **Inefficient for rare disease**

INTERVENTION STUDIES

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- Investigator determines who is exposed, ideally using random methods
- Investigator allocates the exposure and follows for an outcome
- Intervention trials could be done for various purposes:
 - ▣ **Proof of concept trial:** designed solely to produce knowledge about cause and effect, does not test the efficacy of the intervention in actual practice.
 - ▣ **Prevention trial:** interventions are to prevent disease and study participants are persons without disease.
 - ▣ **Clinical trial:** interventions are treatment based on drugs and study participants are persons with disease.

Key Features of Experimental Design

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- **Investigator manipulates the condition under study**
- **Always prospective**

□ **Based on population**

- **Clinical trial** - usually performed in clinical setting and the subjects are patients.
- **Field trial** - used in testing medicine for preventive purpose and the subjects are healthy people. E.g. vaccine trial
- **Community trial**- unit of the study is group of people/community. E.g. fluoridation of water to prevent dental caries.

□ **Based on design**

- **Uncontrolled trial** - no control group. Control will be past experience (history).
- **Non-randomized controlled**- there is control group but allocation into either group is not randomized.
- **Randomized controlled** - there is control group and allocation into either group is randomized.

- Based on objective
 - **Phase I – clinical pharmacologic study**
 - Trial on small subjects to test a new drug with small dosage to determine the toxic effect, safe dose range.
 - Scale: **20-80** healthy individuals
 - **Phase II – efficacy studies**
 - Trial on small group to determine the therapeutic effect
 - Scale: **100-200** patients
 - **Phase III - study on large population to test effectiveness**
 - **Randomized Controlled Trial (RCT)**
 - Often it is multi-centered licensing
 - **Phase IV**
 - Post marketing surveillance
 - Long term prospective assessment of effects & side-effects

INTERVENTION STUDIES...

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- Although experimental studies come in many types, principles are the same and clinical trials dominate the field
- A clinical trial is a prospective study evaluating the effect and value of intervention(s) in human beings under pre-specified conditions.
 - ▣ It is the most definitive tool for evaluation of the applicability of clinical research
- It represents a key research activity with the potential to improve the quality of health care and control costs through careful comparison of alternative treatments.

RANDOMIZED CONTROLLED TRIAL

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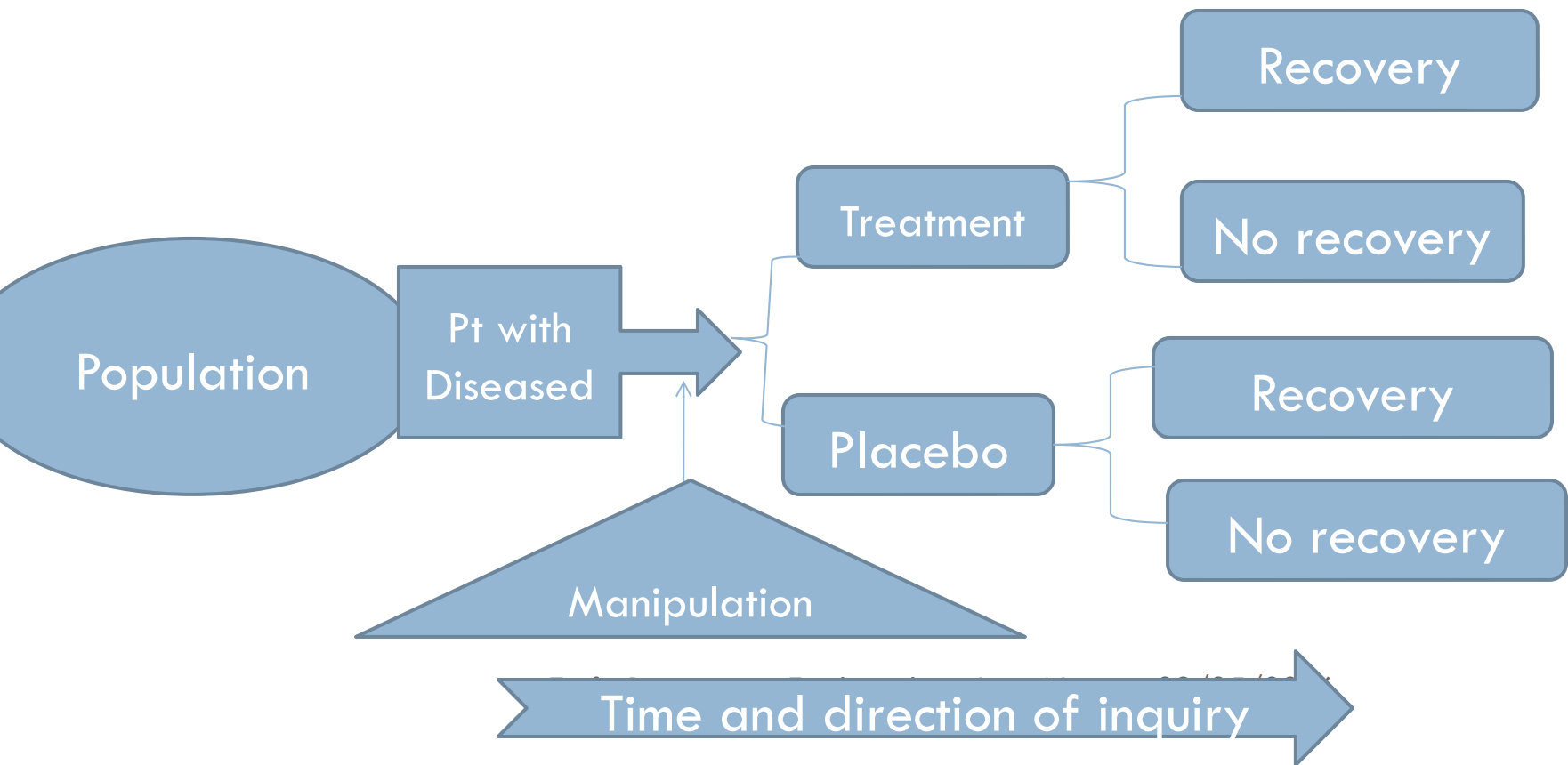
- An investigation involving **intentional** change in some aspect of the status of the subjects, e.g., introduction of a preventive/therapeutic regimen,
 - ▣ It is a design that could bring **high quality of a research** that resembles a controlled experiment **“gold standard”**
- It is similar to cohort study design that individuals are enrolled on the bases of their exposure.
- ***What is the difference b/n cohort and experimental studies?***
 - ▣ Investigator assigns subjects
 - To exposure & non exposure
 - Makes follow up to measure for the occurrence of a disease

- **Based on population it can be classified as**
 - ▣ **THERAPEUTIC** (*secondary prevention*)
 - Interventions made among patients
 - Usually made in health institutions =(clinical trial)
 - ▣ **PREVENTIVE** (*primary prevention*)
 - Interventions performed among healthy people.
 - If individuals are considered as study subjects
 - *field trial*
 - If the community are considered as study subjects
 - community trial

INTERVENTION STUDIES...

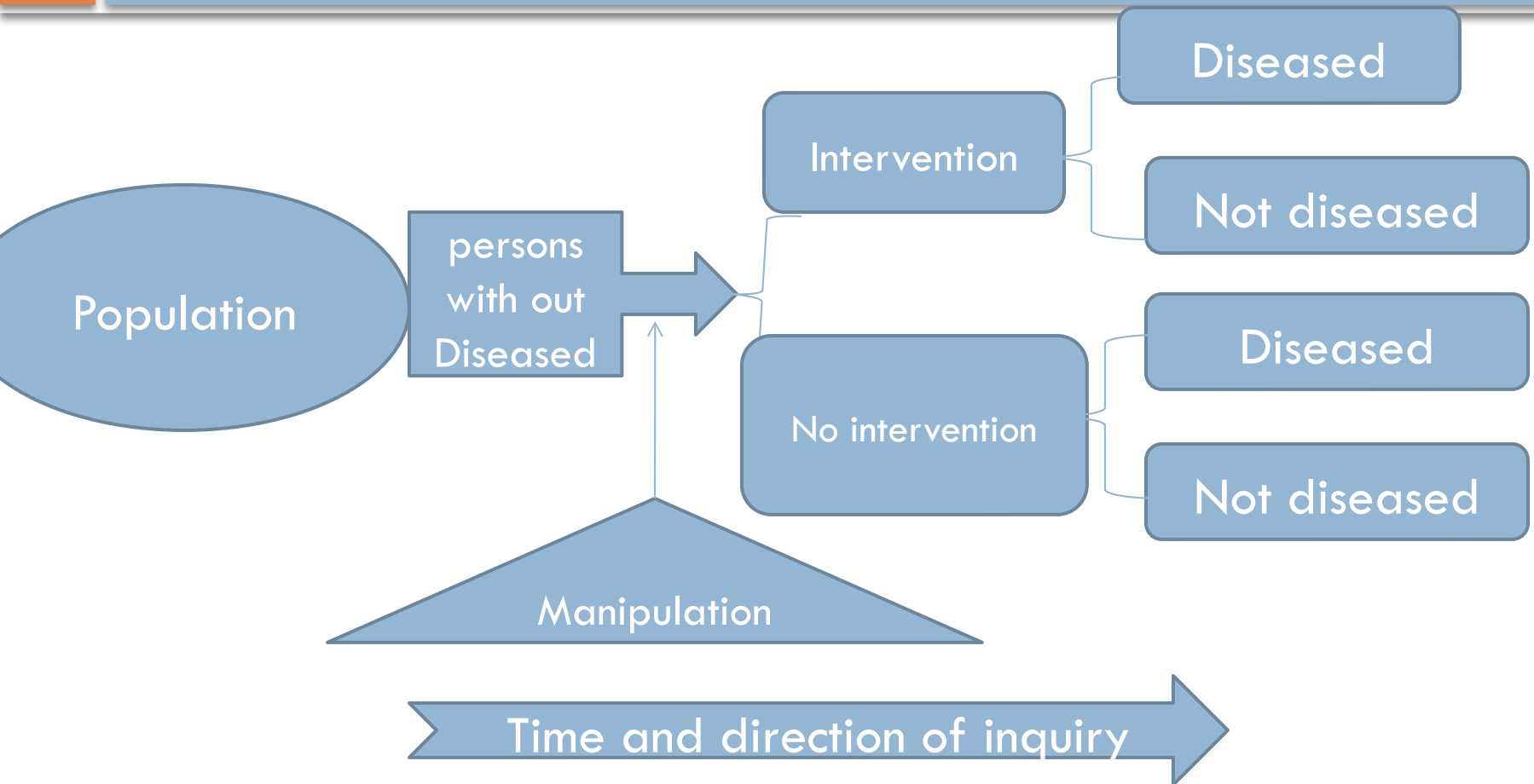
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1. **CLINICAL TRIAL:** (therapeutic trial) A research activity that involves the administration of a test regimen to humans to evaluate its efficacy & safety.



2. FIELD TRIAL

85



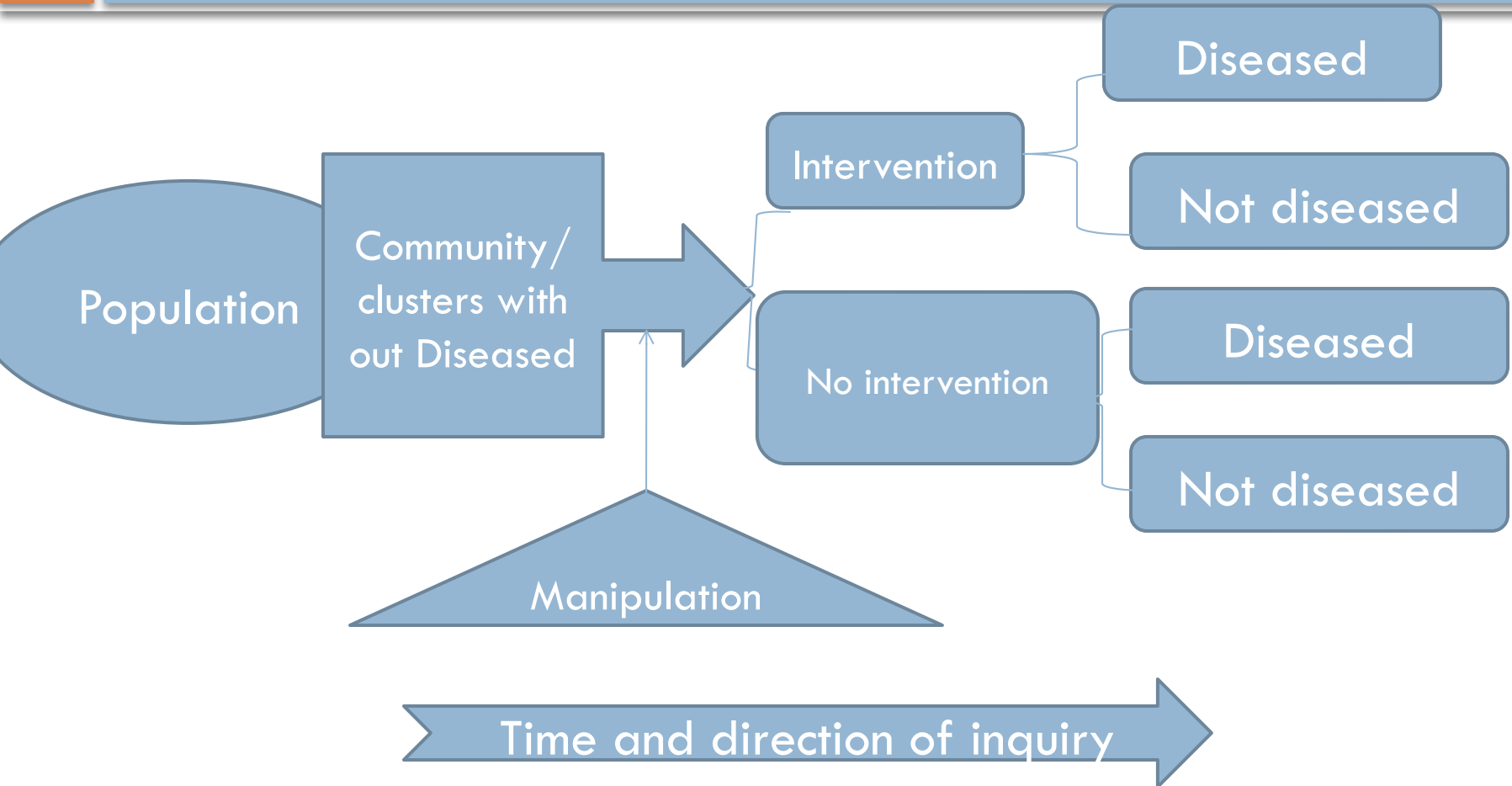
3. COMMUNITY TRIAL

86

- Experiment in which the unit of allocation to receive a preventive or therapeutic regimen is **an entire community or political subdivision**.
- Examples include the trials of fluoridation of drinking water and of heart disease prevention in Ethiopia and Eritrea

COMMUNITY TRIAL

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Selection of Participants

Source population

threats to external validity



Study population



Randomize

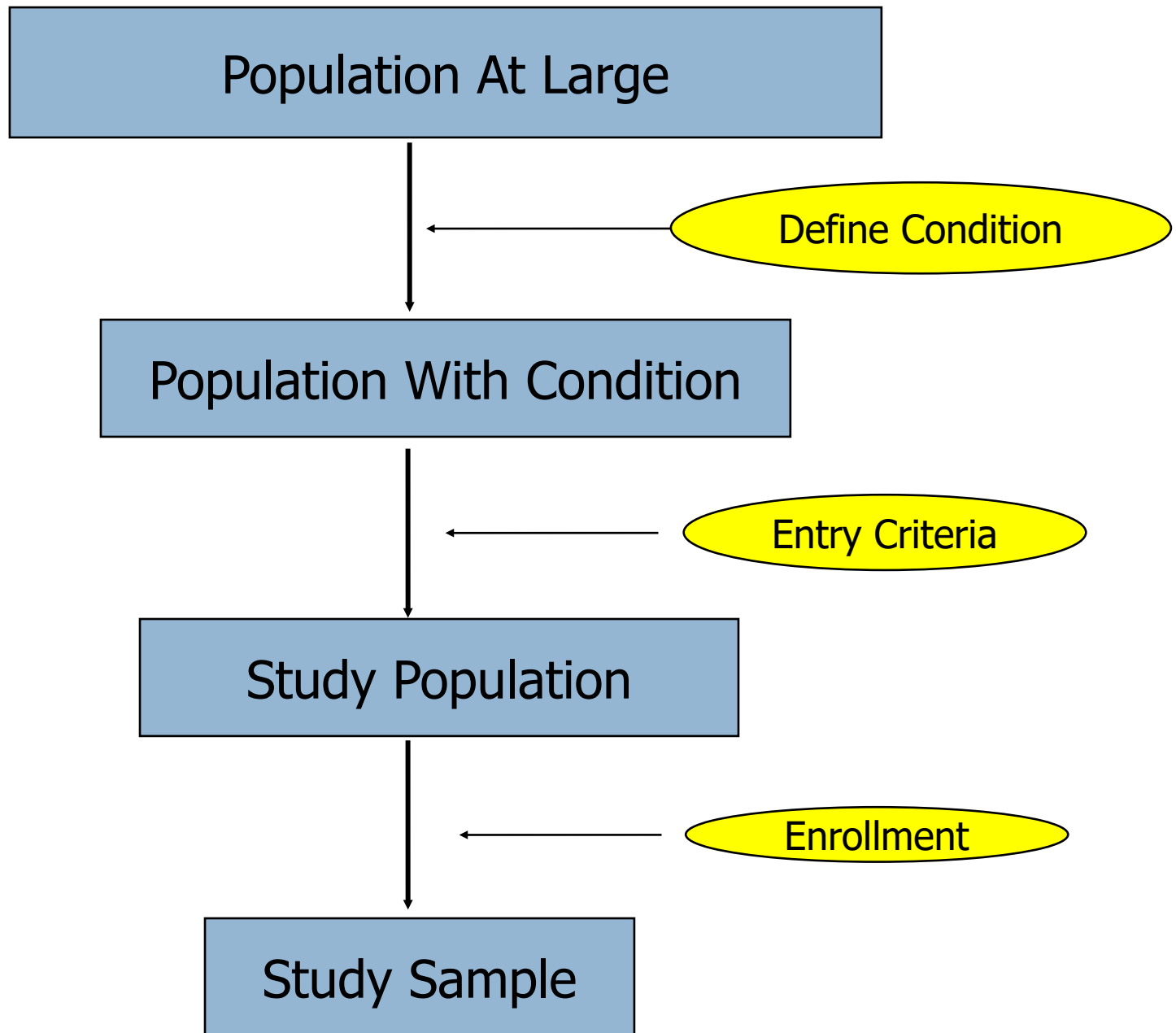


Treatment

No treatment



threats to
internal validity



Sample size

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- ❑ Calculating sample sizes for trials with dichotomous outcomes (eg, sick vs well) requires four components:
 - type I error (α),
 - power,
 - event rate in the treatment group (p_1),
 - RR or event rate in the control group (p_2), $RR = P_1 / p_2$

- **The quality of "gold standard" in intervention studies can be achieved through**
 - ▣ **Randomization**
 - ▣ **Use of placebo**
 - ▣ **Double Blinding**

Experimental.....

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□ **ADVANTAGES**

- Randomization minimize bias & confounding
- Blinding minimizes observer bias
- Placebo could blind patients
 - Ability to demonstrate causal associations when appropriately designed
 - ❖ Use of controlled conditions
 - ❖ Powerful techniques – randomization & blinding
 - Increase subject comparability
 - Decrease potential bias and confounding
 - Investigators control level of exposure
 - ❖ Permits them to establish precise doses or procedures most appropriate to experiment

Disadvantage/weakness

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- Generalizability limited by selection of participants
- Rare/late Side effects of intervention may not be recognize
 - Limited applicability
 - Ethical concerns limit use
 - This is why RCTs focus on interventions that have the potential to reduce rather than increase the risks of morbidity or mortality
 - Bias that can result from differential rates of compliance, withdrawal, or losses to follow-up between experimental and control groups
 - Can alter the magnitude or direction of the relationship between the intervention and study outcome

Disadvantage/weakness

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- ❑ Long-term commitment
- ❑ Considerable expenses necessary to implement and complete a trial
- ❑ Some address narrowly focused questions in artificial settings
- ☞ Applicability to real-world situations may not always be clear
- ❑ Reliance on volunteers and use of strict exclusion criteria
- ☞ Make it difficult to recruit sufficient subjects for a study
- ☞ Can limit external validity of findings
- commercial, political and/or academic pressure

Conclusion...

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Most appropriate study design(s)	
Objective	Study Design
Aetiology or risk factors of disease	Cohort, Case-Control
Extent of disease in population	Survey
Natural history and prognosis of disease	Cohort
Evaluate preventive and therapeutic measures	RCT
Develop public health policy	All

6. Measures of association

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Content

- Strength measure
 - Odds ratio (OR)
 - Relative risk (RR)
- Impact measure
 - Attributable risk
 - Attributable risk percent
 - Population Attributable risk
 - Population Attributable risk percent

Introduction

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- Association is a Statistical relationship between two or more variables
- The strength of association is assessed by Calculating RR,OR or other measures of association.
- Involve comparison of incidence in two or more groups, e.g., exposed and unexposed, to some hazard
- Target group can be either the exposed group or the whole population
- Reference group is the unexposed
- The two incidences can be divided or subtracted

Why do we need them?

- To move from descriptive to analytic epidemiology
- Comparisons within and between populations
- Risk factor identification

Measure of strength...

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- Variables can be related +/- or strong/weak/significant/insignificant.
- Indications of how more or less likely one is to develop disease as compared to another
- **Types of Measures**
 - *Absolute difference*
 - Risk difference
 - Exposed - unexposed
 - *Relative difference*
 - Risk ratios
 - Odds ratios
 - Exposed / unexposed

Measure of strength...

99

- Measures of association between risk factors and disease are often calculated from data presented in a two by two tables:
- **Two-by-two Table Showing Association i.e. r-by-c table**
- Every epidemiologic study can be summarized in a 2-by-2 table.

Exposure	Disease	
	Ill (D+)	Well (D-)
Yes (+)	a	b
No (-)	c	d

Exposure	Disease status		Person years unit
	Yes(+)	No (_)	
Yes	A	-	PY1
No	C	-	PY2

Measure of Strength...

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Key

- A # of individuals who are exposed and have the disease
- B # exposed and do not have disease
- C # not exposed and has the disease
- D # non exposed & do not have the disease
- $A + B$ -Total number of individuals exposed
- $C + D$ -Total number none exposed
- $A + C$ -Total number with disease
- $B + D$ -Total number with out disease
- $A+B+C+D$ -Total number of participant/volunteerr/subject

A. Relative Risk/Risk/Rate Ratio

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The relative risk or risk ratio compares the risk of some health-related event (often disease or death) in two groups, typically in persons exposed to the disease to those not exposed (**used in cohort study**):

The ratio of the risk of an event among the exposed to the risk among the unexposed.

The ratio of the incidence rate in the exposed to the incidence rate in the unexposed

Exposure	Disease		Total
	Yes (+)	No (-)	
Yes (+)	A	B	A+B
No (-)	C	D	C+D
Total	A+C	B+D	A+B+C+D

RR...

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It is a direct measurement of a risk

It is usually used in **Cohort** and **Experimental** study design.

RR = incidence of a disease among exposed
incidence of a disease among non-exposed

□ In short

or $RR = I_e / I_o$

$$RR = \frac{\left(\frac{a}{a + b} \right)}{\left(\frac{c}{c + d} \right)}$$

Strength of association in RR

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- 1.0-----No association(null value)
- 1.1-1.3-----Weak
- 1.4-1.7-----Mild
- 1.8-2-----Moderate
- 3-8 -----Strong
- ❖ For inverse associations (incidence ratio is <1.0).
- ❖ For example: reciprocal of 0.5 is 2.0, which corresponds to a “moderate” association.

Examples-1

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- Among 2390 women aged 16 to 49 years who were free from bacteriuria, 482 were OC users at the initial survey in 1973, while 1908 were not. At a second survey in 1976, 27 of the OC users had developed bacteriuria, as had 77 of the non users.
- Construct a 2 by 2 table, and
- Calculate the measure of association and
- interpret the finding.

B. Odd or Cross Product Ratio

105

- It is an indirect measure of a risk in a disease of rare occurrence (non-proportional ratio).
- It is usually used in a case-control and cross-sectional analytic study designs.
- Calculated as dividing odds of having the disease if exposed to odds of having the disease if non-exposed.
- $OR = \frac{\text{Odds of exposed among cases}}{\text{Odds of exposed among controls}}$
- $OR = a/b/c/d = ad/bc$
- $ODDS\ RATIO = \frac{\text{Ratio of diseased to non-diseased in exposed}}{\text{Ratio of diseased to non-diseased in unexposed}}$

 or
 $= \frac{\text{Ratio of exposed to non-exposed in those with the disease}}{\text{Ratio of exposed to non-exposed in those without the disease}}$

OR...

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Derivation of OR in Cohort study

- $P D+ | E+ = (\text{exposed developed the disease}) = a/(a+b)$
 - $P D- | E+ = (\text{exposed did not develop the disease}) = b/(a+b)$
 - Odds of developing disease among exposed =
 - $\frac{D+ | E+}{D- | E+} = \frac{a/(a+b)}{b/(a+b)} = \underline{a/b}$
 - $P D+ | E- = (\text{non-exposed developed the disease}) = c/(c + d)$
 - $P D- | E- = (\text{non-exposed did not develop the disease}) = d/(c + d)$
 - Odds of developing disease among non-exposed =
 - $\frac{D+ | E-}{D- | E-} = \frac{c/(c+d)}{d/(c + d)} = \underline{c/d}$
 - **Odds ratio** = $a/b : c/d = \underline{ad/bc}$
- RR can be estimated by OR if the following conditions are fulfilled:
- The controls are representative of the general population
 - The selected cases are representative of all cases
 - The disease is rare

Example-3

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- A case control study was conducted on a total 3276 pre-menopausal female nurses to determine using OC as a risk factor for MI. Of 156 women with Myocardial Infarction (MI), 23 were current OC users at the time of their hospital admission. Of the 3120 control women without MI, 304 were current OC users.
- Draw a two by two table
- Calculate the measure of association and
- Interpret it.

Absolute/impact measures

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- It's used to place the association between an exposure and an outcome into a meaningful public health context.
- Whereas a measure of association quantifies the relationship between exposure and disease, and thus begins to provide insight into causal relationships.
- Measures of public health impact reflect the burden that an exposure contributes to the frequency of disease in the population.
- **They all assume (require) a cause-effect relationship exists between the exposure and the outcome.**

Attributable/Excess Risk (AR/ER/RD)

109

- Synonyms: causal rate difference, causal risk difference, Cumulative incidence difference, Incidence density difference
- The proportion of the rate (risk) of a disease or other outcome in exposed individuals that can be attributed to the exposure.
- This measure is estimated by subtracting the rate (risk) of the outcome (usually, incidence or mortality) among the unexposed from the rate (risk) among the exposed individuals.
- $AR = \text{Incidence among exposed (Ie)} - \text{Incidence among non-exposed (Io)}$
- AR quantifies the risk of disease in the exposed group that is attributable to the exposure by removing the risk of disease that occurs due to other causes.

AR/RD...

110

- Attributable Risk: Measure of the potential for prevention of disease if the exposure could be eliminated (assuming a causal relationship).
- It tells us how many cases of disease in exposed people could have been prevented by eliminating the exposure
- Among the EXPOSED: How much of the disease that occurs can be attributed to a certain exposure?
- Attributable Risk: important for
 - Policy decisions
 - Funding decisions (e.g. prevention programs)

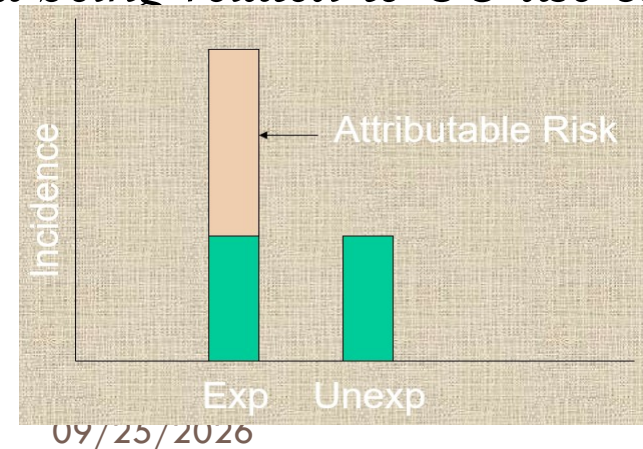
Example-4

111

- From the previous example 1 Calculate the AR and interpret it.

Answers

- **Interpretation:** The excess occurrence of bacteruria among OC users attributable to their OC use is **1566** per 100,000 OC users
- In a population of 100,000 OC users, **5602** would be expected to develop bacteruria, 1566 of those who developed bacteruria being related to OC use & the remainder, 4036, to other factors



Attributable Risk Percent (ARP)

112

- Attributable fraction expressed as a % of the total rate or risk rather than as a proportion.
- Estimates the proportion of the disease among the exposed that is attributable to the exposure, or
- The proportion of the disease in the exposed group that could be prevented by eliminating the exposure

$$AR \% = \frac{(I_e - I_o)}{I_e} \times 100$$

Refer to example 1 and calculate AR%

$$AR \% = \frac{1566/105}{27/482} \times 100 = 27.96 \%$$

- **Interpretation:** If OC use causes bacteruria, about 28 % of bacteruria among women who use OC can be attributed to their OC use and can be eliminated if they did not use oral contraceptives

AR%...

113

- Estimates the proportion of the disease among the exposed that is attributable to the exposure, or
- proportion of the disease in the exposed group that could be prevented by eliminating the exposure

$$AR \% = \frac{(I_e - I_o)}{I_e} \times 100$$

- PAR shows the effect of eliminating the exposure on the population as a whole,
- PAR takes into account not only the actual incidence rate of the outcome but also the prevalence rate of the exposure
- $PAR = AR \times \text{prevalence rate of the exposure}$
- PAR shows the effect of eliminating the exposure on the population as a whole,
- PAR takes into account not only the actual incidence rate of the outcome but also the prevalence rate of the exposure
- $PAR = AR \times \text{prevalence rate of the exposure}$
- **Example**
 - **Cigarette smoking and death from lung cancer.**
 - $AR = 89 \text{ per } 100,000 \text{ per year}$
 - $\text{Prevalence rate of cigarette smoking} = 20 \%$
 - **Calculate PAR.**
 - $PAR = 89 \text{ per } 100,000 \text{ per year} \times 20 \%$
 $= 17.8 \text{ per } 100,000 \text{ per year}$

PAR...Interpretation

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- In a population of 100,000 smokers, 89 deaths from lung cancer per year could have been avoided by preventing them from smoking (this refers to AR).
- In a general population of 100,000 with a prevalence rate of cigarette smoking of 20 %, about 18 deaths from lung cancer per year would be prevented by eliminating cigarette smoking (this refers to PAR).
- Both AR and PAR are used to estimate the effect on disease incidence of eliminating a given risk factor,
- AR estimates reduction in disease incidence only in those exposed,
- PAR estimates reduction in disease incidence in the population as a whole.
- **PAR = Incidence rate in total population minus incidence rate in non-exposed population**

Example-4

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- **Cigarette smoking and death from lung cancer for x population is.**
 - ▣ AR = 89 per 100,000 per year
 - ▣ Prevalence rate of cigarette smoking = 20 %
 - ▣ Calculate PAR.
- **PAR = 89 per 100,000 per year X 20 %**
= 17.8 per 100,000 per year

Interpretation

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- In a population of 100,000 smokers, 89 deaths from lung cancer per year could have been avoided by preventing them from smoking (this refers to AR)
- In a general population of 100,000 with a prevalence rate of cigarette smoking of 20%, about 18 deaths from lung cancer per year would be prevented by eliminating cigarette smoking (this refers to PAR).
- Both AR and PAR are used to estimate the effect on disease incidence of eliminating a given risk factor,
- AR estimates reduction in disease incidence only in those exposed,
- PAR estimates reduction in disease incidence in the population as a whole.
- $PAR = \text{Incidence rate in total population} - \text{incidence rate in non-exposed population}$

Population Attributable Risk Percent (PARP)

118

- PARP: Is the attributable fraction in the population expressed as a percentage.

$$\text{PAR \%} = \frac{\text{PAR}}{\text{Incidence rate in total population}} \times 100$$

Example-5: If PAR = 17.8 per 100,000 per year

Mortality rate in non-smokers = 7 per 105

Mortality rate in the total population = 24.8 per 105 per year

Calculate PAR %

$$\text{PAR \%} = \frac{17.8 \text{ per } 105 \text{ per year}}{24.8 \text{ per } 105 \text{ per year}} \times 100 = 71.8\%$$

Interpretation: 72% of deaths from lung cancer occurring in the general population could be prevented by eliminating cigarette smoking.

Possible Outcomes in Studying the Relationship

119

- No association between exposure and disease
 $AR=0, \quad RR=1$
- Positive association between exposure and disease (more exposure, more disease)
 $AR>0, \quad RR>1$
- Inverse/negative association between exposure and disease
(more exposure, less disease)
 $AR<0 \text{ (negative)}, \quad RR < 1 \text{ (fraction)}$

Validity

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- **Validity:** is a measure of the degree to which something truly represents **what it is being measured/**
- **It's** defined as the extent to which an instrument (for example, a questionnaire or a laboratory test) measures what it is intended to measure.
- Validity can be determined only if there is a reference procedure or 'gold standard': a definitive procedure to determine the characteristic being measured.
 - Ex: information on birth weight obtained from an interview can be validated against hospital records,
 - food-frequency questionnaires against food diaries and biological measurements.
- If the measures are binary – they can be measured by **Specificity, Sensitivity, Positive & negative Predictive values**

Validity...

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Two **types of validity**

- A. Internal validity** - is the degree to which the results of the study are correct for the particular group of people studied
- b. External validity (Generalizability)** - is the extent to which the results of study apply to people not in it.
 - True value – standard or reference value or a theoretical value
 - **Accuracy** – means how close is the measurement with respect to “true” value.
 - **Precision** – is related to the ability of an instrument to measure reproducible values.
 - When talking about the quality of measurement validity types are:
 1. Construct validity
 2. Criterion-related validity
 3. Face and Content validity
 4. Predictive validity
 5. Concurrent validity
 6. Convergent validity
 7. Discriminant validity

1. Construct Validity

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- Evidence is accumulated from a number of procedures when there is no standard method.
- It refers to the fact that the measure captures the major dimensions of the concept under study
- Eg. When we want to measure quality of a service, the instrument should contain all important dimensions of the concept like process, structure, outcome and satisfaction.

2. Face validity

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- When little or nothing is known about the variable being measured, the level of validity obtained is called face validity
- Face validity is the lowest level of validation and is used only when you are beginning to study a particular variable and have no prior research literature to refer to.
- If there is literature on the variable, either theory or research, then face validity is not sufficient.

3. Content Validity

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- Refers to the representativeness of the response categories used to represent each of the dimensions of the concept
- Content validity involves comparing the content of the measurement technique to the known literature on the topic and validating the fact that the tool does represent the literature accurately.
- Content validity is frequently estimated from the review of the literature on the topic or
- Through consultation with experts in the field who have become experts by having done research in the area.
- After you have critically reviewed the literature, you construct your questions or instruments to cover the known content represented in the literature.

4. Criterion-related validity

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- Definition:- Validation procedure involving comparing results with the Gold Standard (you check the performance of your operationalization against some criterion.)

5. Concurrent validity

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- This may be tested by comparing results of one measurement with those of a similar measurement administered to the same population and approximately the same time.
- If both measurements yield similar results, then concurrent validity can be established.

6. Predictive validity

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- It can be examined by comparing the results obtained from the measurement with some actual later-occurring evidence that the measurement aims at predicting
- When there is a high degree of correspondence between the prediction and the actual event then predictive validity is established

Assessing validity of a categorical variable

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- The validity of a categorical variable is carried out by:-
- Sensitivity = proportion of individuals classified as positives by gold standard and who are correctly identified by the study test
$$= a/(a+c)$$
- Specificity = proportion of individuals classified as negatives by gold standard and who are correctly identified by the study test
$$= d/(b+d)$$

Reliability of measurement

129

- **Reliability** is the consistency of measurements, or the degree to which an instrument measures the same way each time it is used under the same condition with the same subjects.
- A measure is considered reliable if a person's score on the same test given twice is similar.
- **Reliability** = repeatability or reproducibility,
 - ▣ Is a measure of the **consistency** of the performance of a test when used under similar circumstances.
 - To be valid, a measurement must be reliable.
 - ▣ However, reliability is not in itself sufficient for validity: in other words, a test may yield the same result consistently, but the result may not be the true (valid) one.
- Poor reliability of a measurement may be due to variation
 - ▣ When a subject is tested on different occasions (biological variation), or
 - ▣ To errors in the measurement technique (observer and instrument variation).

Reliability...

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- **There are three methods of testing or assessing the reliability of research instruments:**
 1. **Test-retest reliability**:-tests for the stability of the instruments (how stable it is over time)
 2. **Inter rater reliability** (tests for equivalence) :- consistency of the results by different investigators or similar tests at the same time
 3. **Split-half reliability** (Internal consistency) :- the measurement of the concept is consistent in all parts of the test.

How do we assess reliability?

131

- **One way is to look simply at percent agreement.**
- **Percent agreement is the proportion of all diagnoses classified the same way by two observers**

Example

132

- Two physicians are each given a set of 100 X-rays to look at independently and asked to judge whether pneumonia is present or absent.
- When both sets of diagnoses are matched, it is found that 95% of the diagnoses are the same

		MD#1	
		Yes	No
MD#2	Yes	43	3
	No	2	52

Is percent agreement good enough?

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- **Do these two physicians exhibit high diagnostic reliability?**
- **Can there be 95% agreement between two observers without really having good reliability?**

Use of the **kappa** statistic to assess reliability

134

- Kappa is a widely used test of inter or intra-observer agreement (or reliability) which **corrects for chance** agreement.
- Cohen's kappa coefficient is a statistic that is used to measure inter-rater reliability for qualitative items.
- It is generally thought to be a more robust measure than simple percent agreement calculation, as K takes into account the possibility of the agreement occurring by chance.

kappa

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It varies from $+1$ to -1

- $+1$ means that the two observers are perfectly reliable. They classify everyone exactly the same way.**
- 0 means there is no relationship at all between the two observer's classifications, above the agreement that would be expected by chance.**
- -1 means the two observers classify exactly the opposite of each other. If one observer says yes, the other always says no.**

Calculating Kappa

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1. Calculate **observed agreement** (cells in which the observers agree/total cells).
2. Calculate **expected agreement** (**chance agreement**) based on the marginal totals
 - **Chance agreement:** Defined as the probability of two raters coding the same behavior at the same time by chance

$$K = \frac{\text{Observed agreement} - \text{Expected Agreement}}{1 - \text{Expected Agreement}}$$

Expected = $\frac{\text{Row total} \times \text{Column total}}{\text{Total sample size (n)}}$, thus $E_{ij} = \frac{R_i \times C_j}{n}$

OBSERVED		MD#2		TOTAL
		YES	NO	
MD#1	YES	43	3	46
	NO	2	52	54
TOTAL		45	55	100

95% AGREEMENT

EXPECTED		MD#2		TOTAL
		YES	NO	
MD#1	YES	20.7	25.3	46
	NO	24.3	29.7	54
TOTAL		45	55	100

50.4 % AGREEMENT

Formula for calculating Kappa from expected agreement

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Observed agreement - Expected Agreement

1 - Expected Agreement

$$\frac{95\% - 50.4\%}{1 - 50.4\%} = \frac{44.6\%}{49.6\%} = .9$$

- You can use this [Cohen's Kappa Calculator](#) to automatically calculate Cohen's Kappa for any two raters.

Guide to use of Kappas in epidemiology and medicine

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- Kappa > 0.80 is considered **Excellent**
- Kappa $0.60 - 0.80$ is considered **Good**
- Kappa $0.40 - 0.50$ is considered **Fair**
- Kappa < 0.40 is considered **Poor or**

Cohen's Kappa	Interpretation
0	No agreement
0.10 - 0.20	Slight agreement
0.21 - 0.40	Fair agreement
0.41 - 0.60	Moderate agreement
0.61 - 0.80	Substantial agreement
0.81 - 0.99	Near perfect agreement
1	Perfect agreement

Measures to improve reliability & validity

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- One is to pilot test your instruments, getting feedback from your respondents.
- Train the interviewers thoroughly.
- All data entry for computer analysis should be "double-punched" and verified.
- Finally, use multiple measures of the same construct if the different measures don't share the same systematic errors,
- Close supervision of the data collectors

Accuracy and Precision

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- *Accuracy* = is the degree of closeness to the truth?
- **Precision** = is the finding due to sampling variation?
 - ✓ Precision in measurement and estimation corresponds to the reduction of random error.
 - ✓ mostly related to sampling variation/error. Solution:
 - ✓ Increase sample size
 - ✓ Improving the efficiency of

The role of chance

- we can draw **inferences** about the experience of an entire population based on **evaluation of only a sample**.
- One of the major problems in drawing such inference is that the play of chance may always affect the results observed simply because of **random variation** from sample to sample.
- One of the major determinants of the degree to which chance affects the findings in any particular study is **sample size**
- The smaller the sample , the more variability there will be in the estimates and the less likely the findings will reflect the experience of the total population.

Role of chance..

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- ❑ The larger the sample on which the estimate is based, the less variability and the more reliable the inference.
- ❑ It is important to quantify the degree to which chance variability may account for the results observed in any individual study.
 - ❑ This is done by performing an appropriate test of statistical significance.
 - ❑ A measure that is often reported from all tests of statistical significance is the P value.
 - ❑ $P \leq 0.05$ - statistically significant.
 - ❑ $P > 0.05$ – no statistically significant association (chance can not be excluded as a likely explanation)

The role of bias

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- **Bias** - is any systematic error in the design, conduct, or analysis of a study that results in a distorted estimate measurement
- The key word in the understanding of the concept of bias is “**different**”.

Two main types of bias:

A. Selection bias

B. Information (Observation) bias

Selection Bias (examples)

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- ☐ **Invitational (who gets invited into the study?)**

- ☐ **Acceptance (who accepted the invitation?)**

 - ☐ Loss to follow-up

 - ☐ Volunteer/Compliance bias

 - ☐ Non-response bias

☐ **Selection bias** is a particular problem in case control and retrospective cohort studies

1. Diagnostic bias

- Diagnostic bias occurs when a disease is more likely to be diagnosed in some one with **exposure to a suspected risk factor**.
 - Happen when knowledge of an exposure is used as a diagnostic criterion
 - For example women who take oral contraceptives (OCs) may be screened more often for breast cancer than women who do not take OCs because of the suspected link between OCs and breast cancer.
 - This would result in breast cancer being diagnosed more readily in those who are exposed to Ocs.

2. Self selection/ Volunteer bias/ Compliance bias

- People who accept to participate in a study, or people who refuse to participate are often quite different from the general population.

3. Non-response bias

- ▣ It reduces the effective sample size, resulting in loss of precision of the survey estimates.

4. Loss to follow up

- ▣ It is major source of bias in cohort & intervention studies
- ▣ it is also a problem in intervention studies before outcome of interest occurs

5. Healthy worker bias

- ▣ Bias in occupational health studies which tend to underestimate the risk associated with an occupation due to the fact that employed people tend to be healthier than the general population

Ways of minimizing selection bias

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- Population-based studies are preferable
- Avoid the inclusions as study subjects of people who have volunteered on their own
- In case-control , select several different control groups.
- In hospital-based case control study, controls are usually selected among patients with disease other than the disease studied.
- keep losses to follow-up to an absolute minimum.
 - For those who are lost, an assessment of as much outcome data as possible.

Ways of minimizing cont...

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- *Loss to follow up*
 - ▣ Calculate estimates of the exposure-disease association assuming the most extreme situations.
 - ▣ Assumptions:
 - all those who were lost to follow up developed the outcome of interest
 - none developed the outcome of interest.
- The results of these calculations provide a range within which the true association will lie.

B. Information bias / Observation bias

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- ❑ Refers to bias which arises during the data collection process
- ❑ It occurs b/c of mistakes in categorizing study subjects with respect to their exposure or disease status

1. Investigator bias/ Interviewer bias/ Observer bias

- ❑ Occurs when investigators collect information differently in different comparison groups

2. Response bias/ Recall bias

- ❑ Occurs as a result of difficulty to recall prior exposures

Information bias cont...

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3. Social desirability bias

- ▣ Occurs because subjects are systematically more likely to provide a socially acceptable response.

4. Placebo effect

- ▣ In experimental studies which are not placebo controlled, observed changes may be ascribed to the positive effect of the subjects belief that the intervention will be beneficial.

5. Hawthorn effect

- ▣ Refers to the change in the dependent variable which may be due to the process of measurement or observation itself

1. **Blinding:** Three types of blinding

- ☐ Single blind
- ☐ Double blind
- ☐ Triple blind

Ways of minimizing

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2. Use of Placebo

- Placebos are **inert treatments intended** to have no effect other than the psychological benefit of offering treatment.
 - Can only be used if there is no accepted treatment for the condition under study.
- Use of placebo minimizes the bias in the ascertainment of both subjective disease outcomes and side effects.
 - It facilitates that both groups in the study gain equal attention.
- **Placebo effect:** tendency for individuals to report favourable response to any therapy regardless of the physiologic efficacy of what they received.

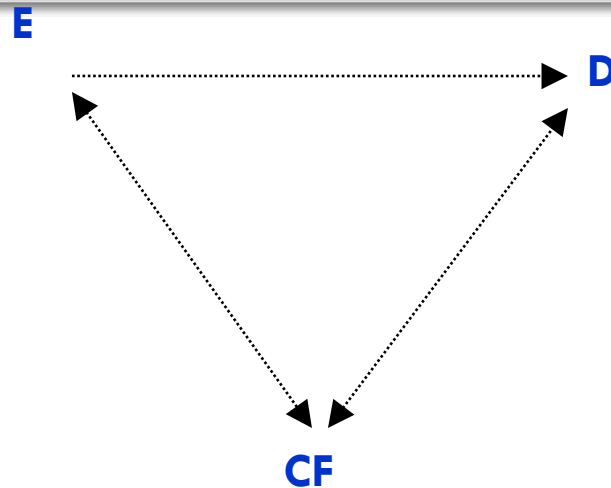
2. **Same standard procedures, instruments, questionnaires, interviewing techniques** should be used for data collection in both comparison groups
3. Classification of study subjects according to their outcome & exposure status should be based on the **most objective & accurate methods available**
4. When exposure status is determined by interview, it should be assessed in several different ways for both groups

- **Confounding** is distortion of the estimated effect of an exposure on an outcome, caused by the presence of an extraneous factor associated both with the exposure and the outcome,
- **Confounding variable, confounder** is a variable that can cause or prevent the outcome of interest, is not an intermediate variable, and is associated with the factor under investigation.
- Confounding may be considered a confusion of effects

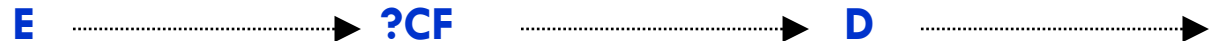
CONFOUNDING

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Confounding IS
present



Confounding NOT
present



Control for Confounding Variables

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☐ In the design:

- ☐ Randomization
- ☐ Restriction
- ☐ Matching (at design and analysis stage)

☐ During analysis:

- ☐ Standardization
- ☐ Stratification
- ☐ Multivariate analysis

9. Screening in disease control

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Contents

- ❑ Definitions and concept of screening
- ❑ Objectives of Screening
- ❑ Principles of Screening in Public health programs
- ❑ Criteria for selection of screening program
- ❑ Types of Screening
- ❑ Criteria and Measures for evaluating Screening programs
- ❑ Public health importance of Se & Sp tests, and prevalence
- ❑ Quality control of screening test
- ❑ Common Biases in screenings
- ❑ Screening program evaluation and monitoring

What is Screening?

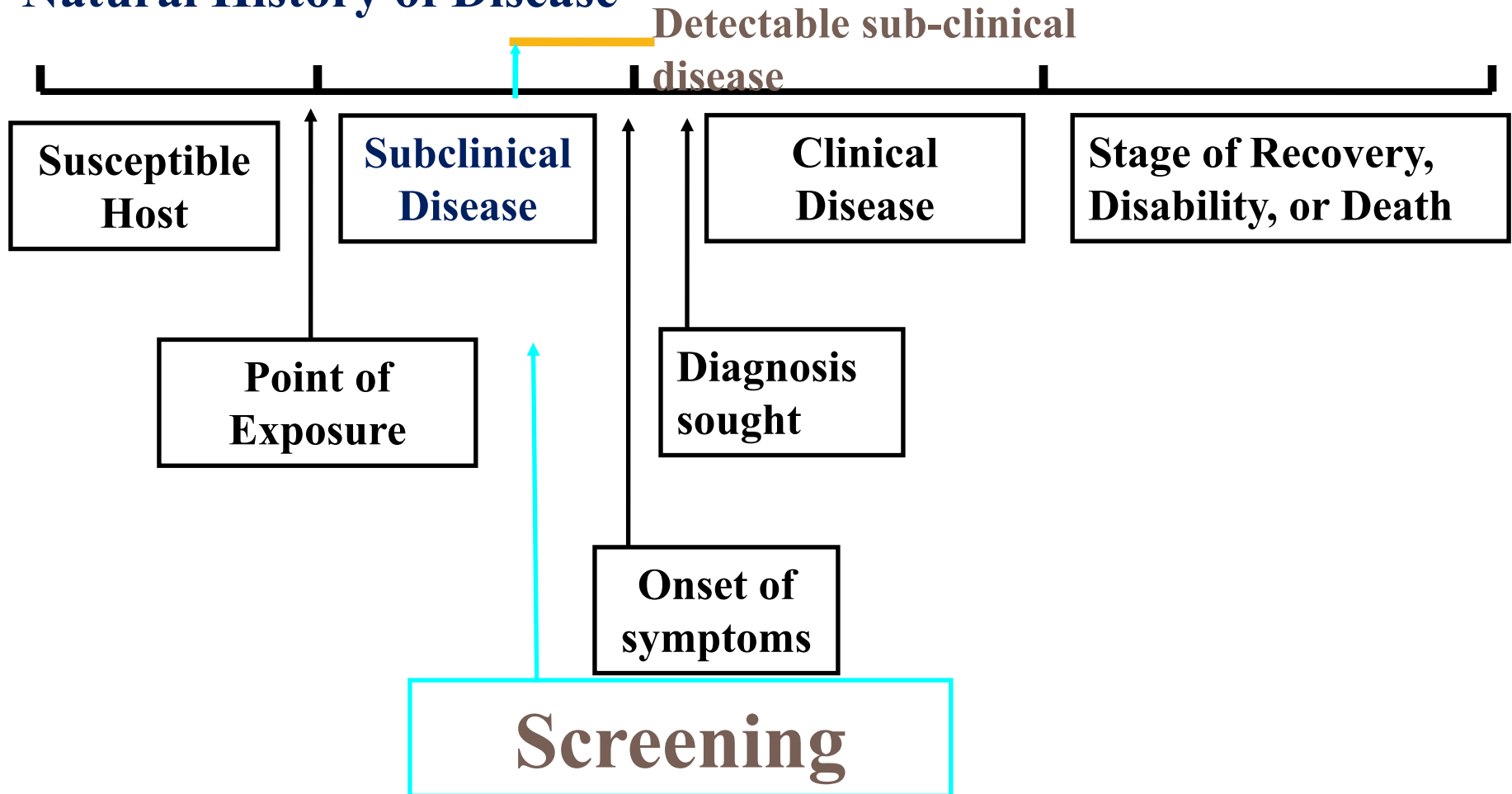
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- Screening is the process to detect, among healthy people, disorders or risk factors of which they are unaware.
- **Screening** is the testing of apparently healthy populations to identify previously undiagnosed diseases or people at high risk of developing a disease.
- **Screening aims** to detect early disease before it becomes symptomatic.
- **Screening** is an important aspect of prevention, but not all diseases are suitable for screening.

Screening...

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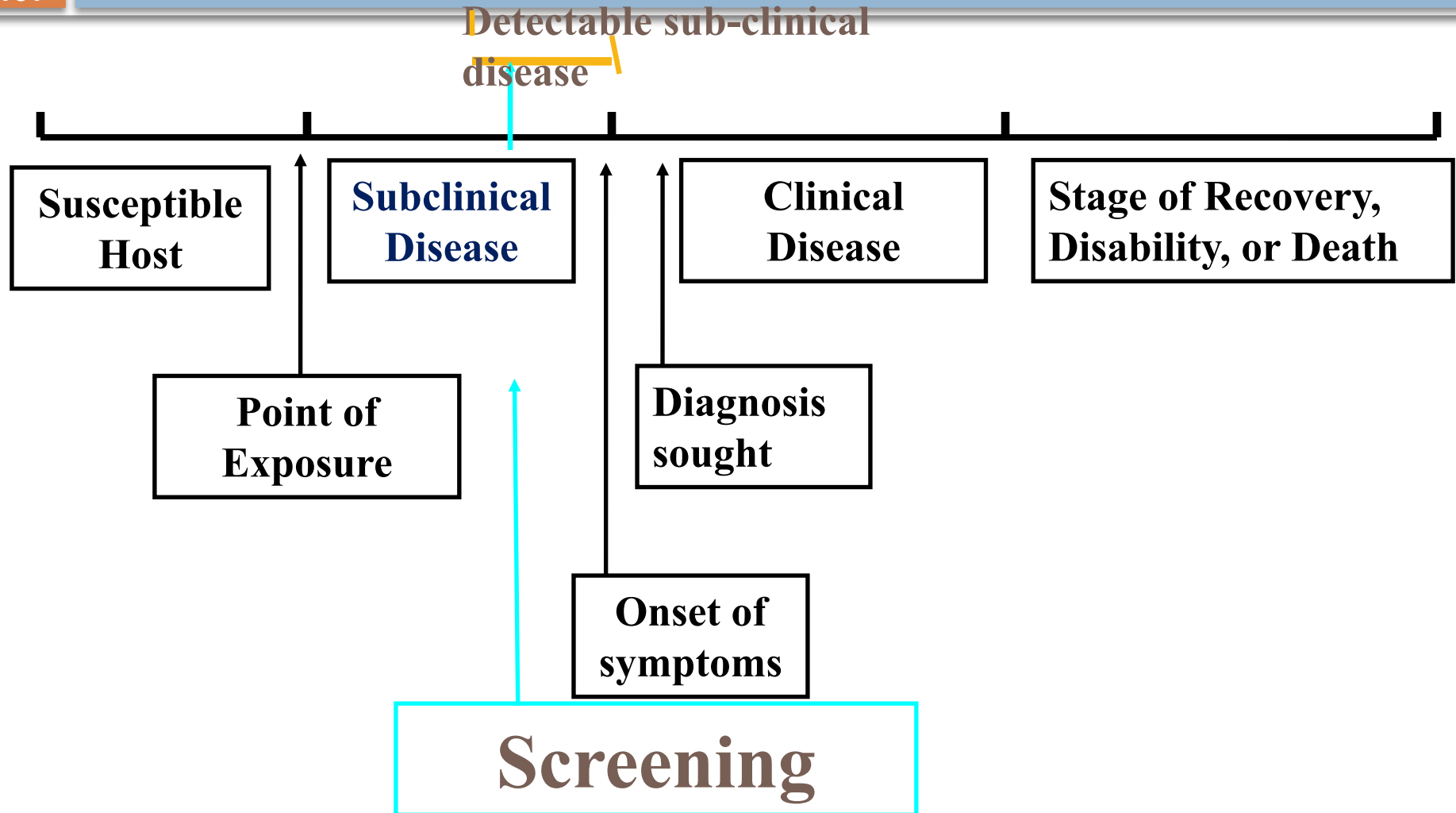
Natural History of Disease



Intro...

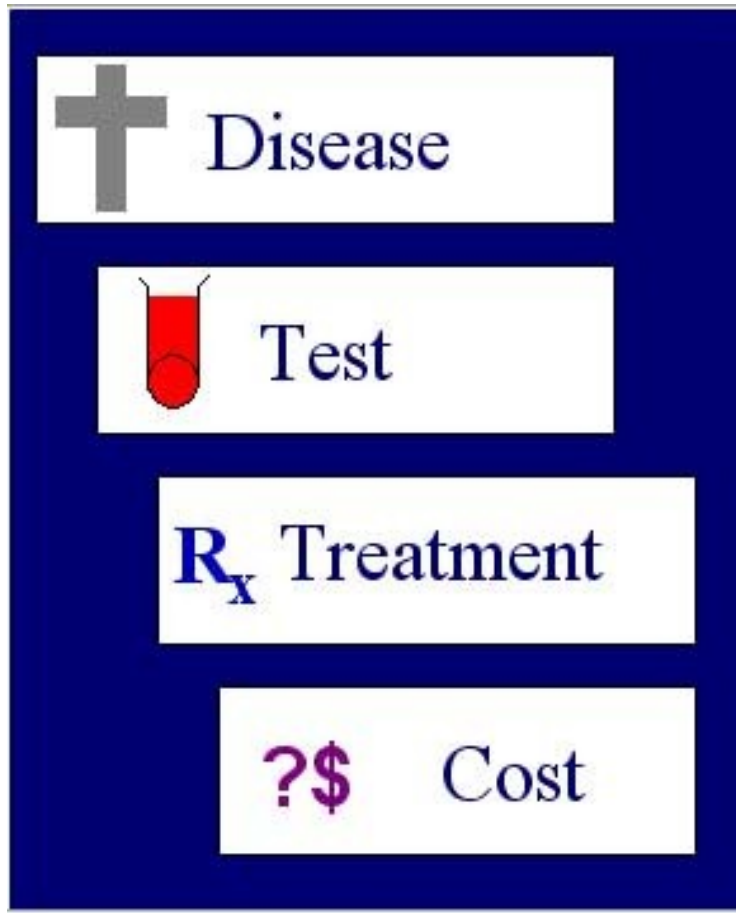
Natural History of Disease

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The Principles of Screening

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- The choice of disease for which to screen;
- The nature of the screening test or tests to be used;
- The availability of a treatment for those found to have the disease;
- The relative costs of the screening.

1. The disease condition

- ▣ Important health condition (prevalent, serious)
 - *Has early stage*
- ▣ High prevalence at asymptomatic stage
- ▣ Adequately understood natural history

2. The test

- ▣ Simple, precise, validated
- ▣ Acceptable to population
- ▣ Agreed diagnostic test(s)
- ▣ Available, adequate, effective and acceptable
- ▣ Sensitive, specific, reliable, simple, cheap, safe

3. The treatment

- ▣ Effective treatment **Available, adequate, and acceptable**
- ▣ Evidence based policies on whom to treat

4. The program

- ▣ Acceptable to public and professionals
- ▣ Benefits outweigh risks
- ▣ Available facilities and staff
- ▣ Overall costs assessed

Characteristics to be evaluated

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□ These characteristics are:

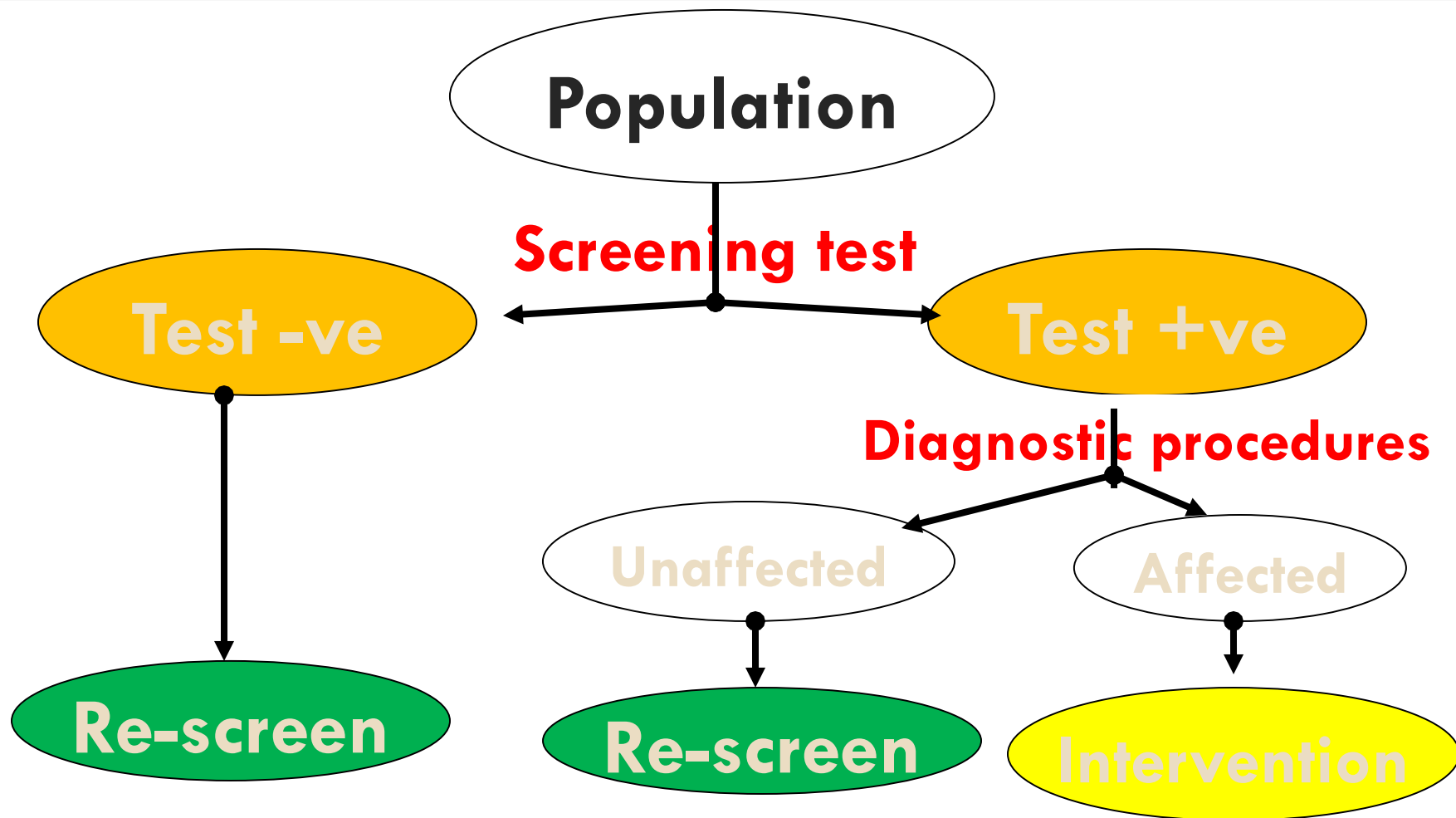
1. Validity
2. Predictive value
3. Reliability
4. Yield

1. **Validity is analogous to accuracy**

- The validity of a screening test is **how well the given screening test reflects** another test of **known greater accuracy**
 - Validity assumes that there is a **gold standard** to which a test can be compared
- **Validity**- does it truly measure what it sets out to measure?
- **Variability/reliability** -does it gives the same measurement each time

Flow diagram for a screening program

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Outcomes of a Screening Test

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Screening Test	True Disease Status		
	Positive (+)	Negative (-)	Total
Positive (+)	A True Positive [TP]	B False Positive [FP]	A+B TP+FP
Negative (-)	C False Negative [FN]	D True Negative [TN]	C+D FN+TN
Total	A+C TP+FN	B+D FP+TN	A+B+C+D TP+FP+FN+TN

Types of screening

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A. Mass

- Large scale screening of **whole population/subgroups** where no selection of population groups is made/irrespective of a particular risk.
- E.g. checking all infants for hearing problems

A. Selective/high-risk/targeted

- For selected **high-risk groups in the population** which may still be large-scale and can be considered as one form of population screening. Example: ANC, PNC

A. Multiple or multiphasic

- Application of ≥ 2 screening test in combination to large groups of people
- Has evolved by combining single screening tests where much time and effort has been spent by population in attending a single test.
- Can be done **sequentially or simultaneously**

D. Case-finding or opportunistic

- Screening of individual coming for other service. E.g. PIHCT

Factors appropriate for screening

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- ◆ Important health problem
- ◆ High prevalence
- ◆ Natural history understood
- ◆ Long latent period
- ◆ Early detection improves prognosis

Evaluation of a screening program

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- ♦ **Reliability**
- ♦ **Feasibility**
- ♦ **Validity**
- ♦ **Performance**
- ♦ **Effectiveness**

Criteria for Evaluating a Screening Test

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- In 1968 WHO proposed the following criteria for screening of diseases

J. M. G. Wilson and *Gunnar Jungner* **classic screening criteria.**

- Includes Validity, Reliability, and Yield

- **Validity**

- ❖ Provide a good indication of who does and does not have disease

- ✓ Sensitivity of the test

- ✓ Specificity of the test

- **Reliability**

- ❖ (Precision): gives consistent results when given to same person under the same conditions

- **Yield**

- ❖ Amount of disease detected in the population, relative to the effort

- ✓ Prevalence of disease/predictive value

Criteria for screening program (WHO, 2008)

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1. The screening program should respond to a recognized need - has potential to decrease in rate of disability /death or both
2. The objectives of screening should be defined at the outset.
3. There should be a defined target population.
4. There should be scientific evidence of screening program effectiveness.
5. The program should integrate education, testing, clinical services and program management.
6. There should be quality assurance, with mechanisms to minimize potential risks of screening.
7. The program should ensure informed choice, confidentiality and respect for autonomy.
8. The program should promote equity and access to screening for the entire target population.
9. Program evaluation should be planned from the outset.
10. The overall benefits of screening should outweigh the harm.

Validity of Screening Test (Accuracy)

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- Sensitivity: Probability to test positive among truly affected.
 - ▣ Is the test detecting true cases of disease? (Ideal is 100%: 100% of cases are detected)
- Specificity: Probability to test negative among truly unaffected.
 - ▣ Is the test excluding those without disease? (Ideal is 100%: 100% of non-cases are negative)

Sensitivity and Specificity

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- $$\text{sensitivity} = \frac{\text{True positives}}{\text{True positives} + \text{false negatives}} =$$

$$\frac{\text{True positives}}{\text{All persons with the disease}} * 100 = \frac{TP}{TP + FN}$$
- $$\text{Specificity} = \frac{\text{True Negative}}{\text{True Negatives} + \text{False Postives}} =$$

$$\frac{\text{True Negatives}}{\text{All persons without the disease}} * 100 = \frac{TN}{TN + FP}$$
- $$\text{Sensitivity} = [a / (a + c)] \times 100$$
- $$\text{Specificity} = [d / (b + d)] \times 100$$

Percent False Negatives and Positives

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- ☐ **Percent False negatives** = % of people with the disease who were not detected by the test.
 - ☐ $PFN = \frac{FN}{FN + TP} * 100$
- ☐ **Percent False Positives** = % of people without the disease who were incorrectly labeled by the test as having the disease.
 - ☐ $PFN = \frac{FP}{FP + TN} * 100$

Predictive Value

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- In the clinical setting, a more important question is:

If the test results are positive (or negative) in a given patient, what is the probability that this patient has (or does not have) the disease?

In other words:

What proportion of patients who test positive (or negative) actually have (or do not have) the disease in question?

Predictive Value...

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- Predictive value is the ability of a test to predict the presence or absence of disease from test results.
- There are two types
 - ▣ Predictive Value of a Positive test(PPV)
 - ▣ Predictive Value of a Negative Test(NPV)
- As prevalence increases, positive predictive value increases.
- As specificity increases, positive predictive value increases.
- As sensitivity increases, positive predictive value also increases, but to a much lesser extent.

Predictive Value...

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□ PPV:- The proportion of diseased individual out of positive test

$$\square PPV = \frac{\text{Diseased individual + ve by the test}}{\text{The whole positive test}} * 100$$

$$\square PPV = \frac{TP}{TP + FP} * 100 = [a / (a + b)] \times 100$$

□ NPV:- The probability of not having the disease when the test is negative.

$$\square NPV = \frac{\text{Non-disease individual -ve by the test}}{\text{The whole negative test}} * 100$$

$$\square NPV = \frac{TN}{TN + FN} * 100 = [d / (c + d)] \times 100$$

Predictive Value...

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$$PPV = \frac{\text{sensitivity} \times \text{prevalence}}{(\text{sensitivity} \times \text{prevalence}) + (1 - \text{specificity}) \times (1 - \text{prevalence})}$$

$$NPV = \frac{\text{specificity} \times (1 - \text{prevalence})}{[(\text{specificity} \times (1 - \text{prevalence}))] + [(1 - \text{sensitivity}) \times \text{prevalence}]}$$

Reliability

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- Reproducibility, repeatability, reliability all mean that the results of a test or measure are identical or closely similar each time it is conducted
- Because of variation in
 - Laboratory procedures,
 - Observers
 - Changing conditions of test subjects
 - **such as time, location , emotions** a test may not consistently yield the same result when repeated
- Different types of variation
 - **Intra-subject variation**
 - **Intra-observer variation**
 - **Inter-observer variation**

Exercise on Screening of diabetes

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- In a hypothetical study, 1000 patients attending a hospital general outpatient department were tested for diabetes using the following two tests: fasting blood sugar (FBS) and glucose tolerance test (GTT). There were 100 patients who had a positive GTT, and they were classified as true cases of diabetes.
- There were also 140 patients with FBS at least 6 mmol/l (cut -off point to distinguish people with diabetes from those who do not have diabetes). Among these 140 patients, only 98 were true cases of diabetes (i.e. only 98 had a positive GTT as well)

Exercise on Screening of diabetes...

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- Based on the above scenario:
- Create a 2 x 2 table, and
- Calculate the sensitivity, specificity and positive and negative predictive values of the FBS in this study population?

Screening Test Result	Disease Status			
		Positive	Negative	Total
	Positive	98	42	140
	Negative	2	858	860
	Total	100	900	1000

Yield

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- Prevalence is the proportion of individuals with the diseases.
 - ▣ $\text{Prevalence} = p(D^+) = (a + c) / n$
- Yield is the proportion of cases detected by the screening program.
- $\text{Yield} = \frac{\text{persons with the disease detected by the test}}{\text{total screened}} \times 100$
- $\text{Yield} = a/n \times 100$

Yield...

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- The **number of cases** of the condition detected by screening test in **relation to the total number of persons screened**

$$Yield = \frac{Tp}{Tn + Tp + Fn + Fp} \times 100\%$$

- Affected by sensitivity, prevalence of unrecognized cases, number of tests employed, frequency of screening and participation of people in screening and follow-up

$$Yield = \frac{\text{persons with the dx detected by the test}}{\text{Total screened}} \times 100\%$$

Biases in screening program

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- Problems exist when comparing survival of screened patients with that of unscreened patients.
- Observational studies and especially survival data are acutely sensitive to:
 - A. Patient self-selection (volunteer) bias
 - B. Lead -time bias
 - C. Length bias
 - D. Over diagnosis bias